

Who Speaks, Which Way, and When?

Simultaneous Speech Translation of Two-Language Conversations

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Abstract

all results are pending

Simultaneous speech translation mostly assumes one speaker and one fixed direction, but a device interpreting between two people who share no language hears both in one mono stream and must decide online who speaks, which way to translate, and when to commit text it cannot retract. We formalise this task, conversational simultaneous speech-to-text translation (CST), and introduce CST-Bench, which renders two-language dialogues as continuous single-channel sessions with natural gaps, translation-mediated gaps or turn-end overlap. It pairs 86 German–English telephone dialogues (real speech, synthetic timing, 640 human-translated turns), released as deterministic build scripts, with an evaluation-only synthetic English–German and Korean–English track. Its scorer, tested by controlled perturbations, reports per-direction quality, within-turn lag, the listener’s wait after each turn, switch latency and wrong-direction output, plus speaker-attributed transcription and diarization errors; Kendall’s τ between system rankings on synthetic and real sessions is [TBD]. Without oracle segmentation, the best open pipeline loses [TBD] COMET ([TBD] to turn segmentation, [TBD] to routing) and, under turn-end overlap, emits [TBD]% wrong-direction words. Our open streaming speech LLM, MODEL, writes speaker-tagged transcripts and direction-tagged translations into one token stream on an 80 ms clock and learns to commit translations at alignment-verified meaning units; without oracle input, it recovers [TBD]% of this loss at a median wait of [TBD] s. *[Draft: planned; the synthetic track, the overlap renders, most scorer modules and MODEL beyond its streaming ASR backbone are not built yet, and*

1. Introduction

Two people who share no language can talk through a device that translates both sides, a goal of early speech translation systems (Waibel, 1996; Wahlster, 2000) that deployed services now offer (Lewis, 2015). The device stands in for a dialogue interpreter, who coordinates turns as well as translating them (Wadensjö, 1998; Roy, 2000). Simultaneous speech translation (SimulST) has scaled to large streaming models (Seamless Communication et al., 2023; Labiausse et al., 2025), which IWSLT evaluates on long unsegmented audio (Abdulummin et al., 2025; Adelani et al., 2026). Yet most work uses human pre-segmented speech (Papi et al., 2025), and SimulST benchmarks translate one source language in one fixed direction, mostly from one speaker.

Conversation changes the problem in four ways (Figure 1). (i) *The direction is latent*: the system must segment one mono stream at speaker changes and route each turn to its direction; with one language per speaker, routing is language identification, but changes must be detected within gaps of a few hundred milliseconds or during overlap. (ii) *Turns are short and close*: the median silence-trimmed turn in our real dialogues lasts 2.86 s, and in natural conversation replies follow gaps of about 200 ms (Levinson & Torreira, 2015) or overlap, as about 40% of speaker changes do (Heldner & Edlund, 2010). (iii) *Output is committed*: a listener may reply once text appears, and no revision undoes that reply, so we score append-only output; re-translation systems (Arivazhagan et al., 2020) take part through a commit rule (Section 6.1). (iv) *Latency is felt at turn ends*: listeners wait after each turn (EndOffset) and at each change of direction (switch latency), beyond the within-turn lag that StreamLAAL measures (Papi et al., 2024). We therefore pose conversational simultaneous speech-to-text translation (CST; Section 3) as streaming structured prediction from a mixture, with a latent speaker and direction state and append-only, time-stamped translations whose language encodes the direction, and score it under realistic and oracle segmentation. Interruptions, barge-in, backchannels, more speakers, code-switching and speech output are out

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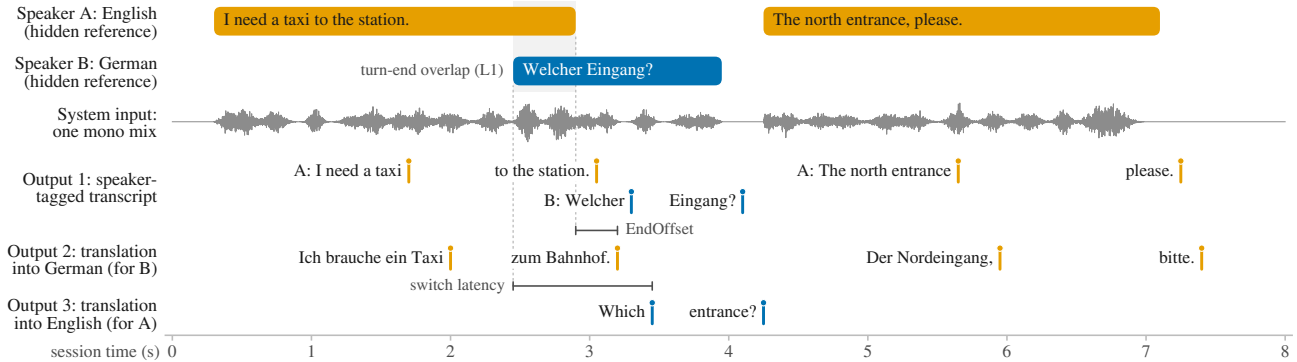


Figure 1. CST on one mono stream (invented utterances). The system hears only the mix and commits, without retraction, translations into each listener’s language and, optionally, a speaker-tagged transcript (Output 1). Ticks: committed pieces; the shaded band is a turn-end overlap (condition L1, Section 4). EndOffset: from a turn’s end to its last translated word; switch latency: from the start of a turn in a new direction to its first translated word.

of scope.

Among the closest systems (Table 1), JSTAR (Moritz et al., 2025) jointly streams recognition and translation of bilingual conversations and infers the speaker, hence the direction, but from spatial cues of multi-microphone smart glasses, on in-house data. DiariST (Yang et al., 2024) and STAC-ST (Zuluaga-Gomez et al., 2023) attribute speakers in one channel but translate from one source language, STAC-ST only offline. Seed LiveInterpret 2.0 (Cheng et al., 2025) names multi-speaker confusion as a challenge, but this closed system is evaluated per direction without overlap and only for Chinese–English, absent from CST-Bench, so we do not run it. Neither single-channel input nor conversational translation is new, but to our knowledge no public benchmark, open baseline or model addresses streaming translation of both sides of a two-language conversation from one mono stream, where turns and their directions are latent and may overlap.

We ask how much of the difficulty lies in segmentation and routing rather than translation (RQ1), how quality, lag and listener wait trade off as turn-taking tightens and for a distant language pair (RQ2), whether one streaming model closes the oracle–realistic segmentation gap without oracle input (RQ3), and whether synthetic sessions rank systems as real ones do (RQ4). Beyond the task, we contribute:

- **CST-Bench** (Section 4): 86 German–English telephone dialogues (Bavarian Archive for Speech Signals (BAS), 2001; Reichel et al., 2016) with real speech and 640 human-translated turns (CST-Bench-TAXI) and a synthetic En↔De and Ko↔En track, rendered with natural gaps, translation-mediated gaps (on CST-Bench-TAXI) or turn-end overlap. [Draft: planned; the synthetic track (CST-Bench-Syn) and the L1 renders are not built yet]

- **A tested scorer** (Section 4) of per-direction quality, wrong-direction output, StreamLAAL, EndOffset and switch latency, plus speaker-attributed WER (cp-WER) and diarization error rate (DER). [Draft: partly planned; v0.1 computes BLEU, chrF, StreamLAAL, EndOffset, empty turns and wrong-language words given turn IDs; only two perturbation rows exist, as preliminary runs, and the untranslated-source row exposes a StreamLAAL artefact (Table 10)]
- **An oracle-component factorial** (Section 6) of open systems over segmentation, routing and overlap. [Draft: planned; only the offline systems on oracle turns have been run, as preliminary runs]
- **MODEL** (Section 5): an open streaming speech LLM writing speaker-tagged transcripts and direction-tagged translations into one token stream and committing at alignment-verified meaning units; a decoupled twin and a same-backbone cascade test whether this coupling helps. [Draft: planned; only the streaming ASR backbone is built]

2. Related Work

Simultaneous translation. Read/write policies range from wait- k to local agreement and attention-guided emission (Ma et al., 2019; Liu et al., 2020; Papi et al., 2023b), and segmentation is learned inside the model (Zhang & Feng, 2023) or handled for unbounded audio (StreamST; Papi et al., 2024). These methods, like StreamSpeech, SeamlessStreaming and Hibiki(-Zero) (Zhang et al., 2024; Seamless Communication et al., 2023; Labiausse et al., 2025; 2026), assume one speaker and one direction, as do SimulST toolkits and latency metrics (Ma et al., 2020a; Gaido et al., 2025; Polák et al., 2025); our scorer, `cstbench eval`, instead resegments per output language, reports SimulEval’s EndOffset per turn and StartOff-

Table 1. CST vs. the closest settings and systems (sources: Appendix I). Spk×Lang: speakers×languages; Auto dir.: system picks the direction; Ovl.: overlap evaluated (sim.: simulated); Str.: streaming; Out.: A transcript, T translation, D speakers or turns, S speech; o partial; n.r. not reported. ^aTAXI rebuilt by our scripts.

Work	Input	Spk×Lang	Dir.	Auto dir.	Ovl.	Str.	Out.	Pub. data	Open wts
SimulST/StreamST	mono	1×1	X→Y	×	×	✓	T	✓	o
SeamlessStreaming	mono	1×1	X→Y	×	×	✓	T,S	✓	✓
StreamSpeech	mono	1×1	X→En	×	×	✓	A,T,S	✓	✓
Hibiki(-Zero)	mono	1×1	X→En	×	×	✓	T,S	✓	✓
STAC-ST	mono	2×1	Es→En	×	✓	×	A,T,D	✓	×
DiariST	mono	N×1	Zh→En	×	✓	✓	T,D	✓	×
JSTAR	array	2×2	Es↔En	✓	sim.	✓	A,T,D	×	×
Seed LiveInterpret 2.0	n.r.	o	Zh↔En	o	×	✓	T,S	✓	×
CST-Bench + MODEL (ours)	mono	2×2	En↔De Ko↔En	✓	sim.	✓	A,T,D	✓ ^a	✓

[Draft: planned in our row: the L1 overlap renders, CST-Bench-Syn (incl. Ko↔En), transcript and diarization scoring, and MODEL with its weights; the TAXI build scripts and the translation scorer exist.]

set at direction changes (switch latency), and counts wrong-direction words (Section 4). [Draft: planned in *cstbench eval*: switch latency and the wrong-direction detector (Appendix C)]

Speakers and turns. Serialized output training transcribes overlapping speakers in one sequence, also in time order (t-SOT) and with speaker attribution (Kanda et al., 2020b; 2022a;b). DiariST extends t-SOT to speaker-attributed streaming translation (Yang et al., 2024), and STAC-ST merges the two sides of Fisher-CALLHOME calls into one channel (Zuluaga-Gomez et al., 2023). Turn-taking models predict both speakers’ voice activity but learn from monolingual dyads (Ekstedt & Skantze, 2022; Inoue et al., 2024).

Conversational data. Of the corpora in Table 8, only TAXI (Section 4) combines real two-language dialogues, human translations of both sides and free research access; others are one-sided (MSLT; Federmann & Lewis, 2016), re-enacted (DRAL; Ward et al., 2024), machine-mediated (InCroMin; Čechovič et al., 2025) or from one source language (Fisher-CALLHOME; Post et al., 2013). Appendix I extends this section.

3. Conversational Simultaneous Translation

Definition 1 (CST). A session x on $[0, T]$ is mono audio of one dialogue between speakers A and B with languages $\ell_A \neq \ell_B$. Its turns n , hidden from the system, have speaker s_n , interval $[b_n, e_n]$ (turns may overlap), transcript in ℓ_{s_n} and reference translation r_n . Given only the language pair, a system reads x as a stream and appends committed pieces (t_i, λ_i, y_i) : text y_i in language λ_i , which fixes the direction, emitted at time t_i (on the ideal clock, the audio consumed at

Table 2. CST-Bench composition. Drawn gaps exclude up to 0.2 s of turn margins. *Planned. [‡]Korean MT. [§]Partly post-edited MT.

	TAXI (real speech)			Syn (TTS)*	
	L0 nat.	L0 med.	L1*	En↔De	Ko↔En
References	human			human [§]	mixed [‡]
Sessions / turns	86 / 640				
Speech (min)	40.378				
Median gap (s)	0.223	2.026			
Overlap ratio	0.0	0.0			

commitment). Optional channels carry a speaker-attributed transcript and speaker activity. Outputs depend only on x before their emission time and are never revised.

This is the main, *realistic* protocol; the *oracle-segmentation* one, a diagnostic, not a bound (Appendix D), reveals gold turn intervals and languages. Both use one resegmentation (Section 4); gold turn IDs serve diagnostics only. Turn timing ignores system output, so L0 natural gaps and the overlap stress test L1 (Section 4) assume instant translation. [Draft: L1 is planned]

CST as sequential prediction. At step k , a policy π reads audio $X_{\leq k}$ and its committed output H_k , updates a latent speaker, direction and turn state S_k , and commits translation and transcript tokens Y_k, Z_k , if any. We compare policies by quality at a lag budget, $\max_{\pi} Q(\pi)$ s.t. $\text{StreamLAAL}(\pi) \leq \Lambda$. A cascade factorises π : a segmenter and router fix $\hat{S}_k$ without reading the translation, and a translator commits under $P(Y_k | \hat{S}_k, X_{\leq k}, H_k)$. We hypothesise that S_k and Y_k are better decided jointly, as lexical completion helps predict turn ends (Ekstedt & Skantze, 2020), which call for commits; a decoupled twin and a same-backbone cascade test this (Section 6.5).

4. CST-Bench

CST-Bench renders two-language dialogues as continuous mono sessions that differ only in turn-taking, and scores what a system commits, and when (Figure 2 and Table 2). It serves evaluation only; nothing is tuned on CST-Bench-TAXI.

Real track: CST-Bench-TAXI. TAXI 2.5 holds telephone dialogues between German-speaking taxi dispatchers and English-speaking clients, with human translations of both sides (Bavarian Archive for Speech Signals (BAS), 2001; Reichel et al., 2016). Push-to-talk recording left one file per turn, without timing or overlap, so in CST-Bench-TAXI only the timing is synthetic, and the mono mix of its 640 usable turns simulates a shared device, not the original calls (Appendix A). Language coincides with role, and the German dispatcher opens 85 of 86 sessions.

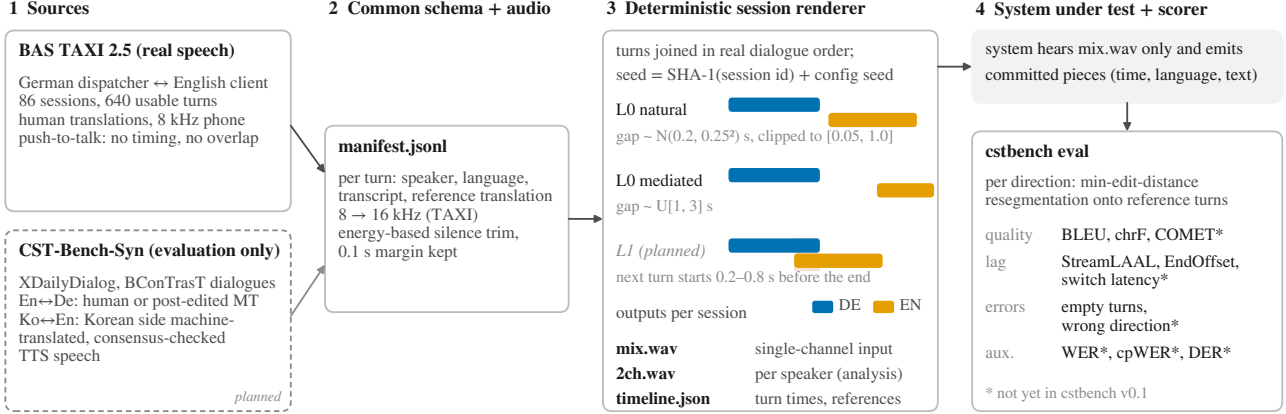


Figure 2. The CST-Bench pipeline (planned parts dashed or starred). For TAXI, we release only code, checksums and per-turn scores.

Synthetic track: CST-Bench-Syn. [Draft: planned: nothing in this track is built yet] Voiced by TTS, it cannot replace a real test set but adds overlap measurable at the word level, a distant pair (Ko↔En) and both speaker–language assignments, so that language predicts neither role nor first direction. Speakers take turns from two language versions of one XDailyDialog (Liu et al., 2023) or BConTrasT (Farajian et al., 2020) dialogue, each supplying the other’s references; Korean text is consensus-filtered machine translation (Appendix B).

Timing conditions. Turns keep their dialogue order, with within-speaker pauses from $\mathcal{U}[0.3, 0.8]$ s and per-session seeds (Appendix A). At speaker changes, *L0 natural* draws gaps from $\mathcal{N}(0.2, 0.25^2)$ s clipped to [0.05, 1] s, around the typical 200 ms gap, and *L0 mediated* from $\mathcal{U}[1, 3]$ s, as when listeners first read the translation; as gaps separate turn intervals that keep 0.1 s margins, silences between speech are up to 0.2 s longer. *L1* starts each new speaker’s turn $\mathcal{U}[0.2, 0.8]$ s before the previous one ends, adding overlap to real speech. Per-speaker channels serve an oracle-separation diagnostic. [Draft: planned: *L1*, with a cap for short turns fixed before any baseline run, and word-level overlap; decision needed: whether the *L1* overlap is measured between energetic frames or between margin-inclusive intervals, which leaves up to 0.2 s less overlapping speech]

Scoring. `cstbench eval` resegments the committed words of each session and output language onto the references of that direction by word-level Levenshtein alignment (Matusov et al., 2005), keeping each word’s emission time, and reports COMET (Rei et al., 2022a), chrF (Popović, 2015) and BLEU (Papineni et al., 2002) per direction on lowercased, unpunctuated text (Appendix C). For turn n , with $L_n = e_n - b_n$ and words $j = 1, \dots, |h_n|$ resegmented to it and emitted at $t_{n,j}$, let $d_{n,j} = \max(0, t_{n,j} - b_n)$

and $\rho_n = L_n / \max(|h_n|, |r_n|)$. Then

$$\text{StreamLAAL}_n = \frac{1}{J_n} \sum_{j=1}^{J_n} (d_{n,j} - (j-1) \rho_n), \quad (1)$$

$$\text{EndOffset}_n = t_{n,|h_n|} - e_n, \quad (2)$$

$$\text{SwitchLatency}_n = d_{n,1} \quad \text{for } n > 1, s_n \neq s_{n-1}, \quad (3)$$

where J_n is the first j with $d_{n,j} \geq L_n$ (else $|h_n|$). Equation (1) is length-adaptive Average Lagging (LAAL; Ma et al., 2019; Papi et al., 2022) on resegmented turns, i.e., StreamLAAL (Papi et al., 2024): the lag within a turn, L_n for output committed at turn ends. As StreamLAAL ignores words after J_n , EndOffset, SimulEval’s EndOffset taken per turn (Ma et al., 2020a; Seamless Communication et al., 2023), measures the wait after the speaker stops; switch latency is the matching StartOffset at direction changes. A committed piece is *wrong-direction* if turns are open but none is translated into its language, if a text language identifier assigns it to the other language, or if it copies an open turn’s transcript; we report the share of committed words in such pieces. Transcripts are scored by WER and cpWER (Watanabe et al., 2020; for Korean, cpCER on characters), speakers by DER.

Statistics and validation. Comparisons carry 95% confidence intervals (CIs) from a session-clustered paired bootstrap (Koehn, 2004); main comparisons, fixed before the test runs, are Holm-corrected (Holm, 1979; Appendix H). Controlled perturbations of the reference output test whether BLEU, chrF, StreamLAAL, EndOffset and wrong-direction words move only as expected (Table 10). Copying the source lowered De→En StreamLAAL below that of the reference committed at turn ends ($2.276^\dagger$ vs. $3.086^\dagger$ s; † preliminary), so Table 3 shows latency next to quality and empty turns, and Appendix C defines a matched-word latency. [Draft: scorer v0.1 computes BLEU, chrF, StreamLAAL, EndOffset, empty turns and wrong-language words given gold turn IDs; the other metrics (including the length

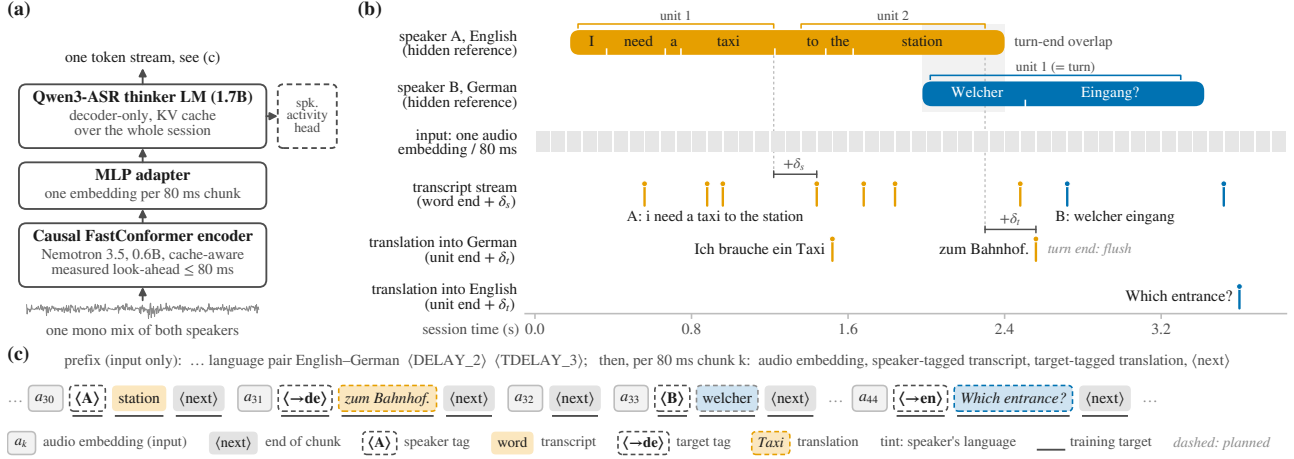


Figure 3. MODEL and conversational delayed streams (illustrative timings; $\delta_s = 2$, $\delta_t = 3$). (a) Architecture. (b) Targets: a transcript word is due δ_s chunks, the translation of a verified meaning unit δ_t chunks, after the chunk in which it ends; a turn end forces a commit. (c) The resulting token stream. Dashed: planned. [Draft: built today: the encoder–adapter–LM backbone, trained for single-speaker English and Korean transcription]

ratio and the matched-word latency), the bootstrap and a pinned number normaliser are being added; rows 3–10 of Table 10 are planned]

5. MODEL: Conversational Delayed Streams

After each $\Delta = 80$ ms audio chunk, the causal speech LLM MODEL writes zero or more tokens of a speaker-tagged transcript and translations into both languages (Figure 3): reading and writing become next-token prediction. Unlike delayed streams modeling (one delayed text stream; Zeghidour et al., 2025), conversational delayed streams delay each channel separately; unlike Hibiki’s alignment-based per-word delays (Labiausse et al., 2025), they commit units verified by translating the source heard so far. [Draft: planned: only the backbone of Section 5.1 is built; Sections 5.2 to 5.4 describe the design, and no CST result exists yet]

5.1. Streaming backbone

A cache-aware FastConformer encoder from Nemotron 3.5 ASR Streaming 0.6B (NVIDIA, 2026), with a measured look-ahead of at most 80 ms, feeds one embedding a_k per chunk k , via an MLP adapter, to the 1.7B Qwen3-ASR language model, the *thinker* (Shi et al., 2026), replacing its non-causal audio encoder (Appendix F). After a prefix with the language and a delay token $\langle \text{DELAY}_\delta \rangle$, each chunk holds a_k , the text tokens due in it and an end-of-chunk token $\langle \text{next} \rangle$; a word ending at t is due in chunk $\lceil t/\Delta \rceil + \delta$.

5.2. One serialized stream

The prefix names the language pair and two delays, δ_s for transcripts and δ_t for translations, sampled per training ses-

sion. Chunk k reads

$$B_k = a_k \langle \hat{s}_1 \rangle z_1 \cdots \langle \rightarrow \lambda_1 \rangle y_1 \cdots \langle \text{next} \rangle, \quad (4)$$

where z_j are words of speaker $\hat{s}_j \in \{A, B\}$, numbered by arrival, and y_i translates one unit into language λ_i . Each tagged segment is a committed piece (Definition 1), making the speaker and direction state explicit; under overlap, words are ordered by end time, as in t-SOT (Kanda et al., 2022a) and JSTAR (Moritz et al., 2025). We minimise

$$\mathcal{L} = - \sum_{m \in \mathcal{Y}} w_m \log p_\theta(o_m | o_{<m}) / \sum_{m \in \mathcal{Y}} w_m \quad (5)$$

over the labelled positions $\mathcal{Y}$ (text, tags, $\langle \text{next} \rangle$) of the tokens o , with $w_m < 1$ only for $\langle \text{next} \rangle$ (Appendix F); a two-slot, arrival-order activity head (Park et al., 2025) adds a binary cross-entropy and serves DER. The session-long KV cache lets both sides condition every decision. [Draft: planned: the two speaker tags are reserved in the vocabulary but untrained; translation segments, the translation delay token and the activity head do not exist yet]

5.3. Commit targets and latency

A teacher LLM (Qwen3.8-27B) translates each source turn $u_{1:I}$ into v , and contextual alignment (Labiausse et al., 2025) maps target word v_j to $\alpha_j = \arg \max_i [\phi_j(i) - \phi_j(i-1)]$, $\phi_j(i)$ being its log-probability when v is force-decoded from $u_{1:i}$. Position i is a candidate if $\{j : \alpha_j \leq i\} = \{1, \dots, \kappa\}$ with κ above the committed length, as meaning units require (Zhang et al., 2020), and the source has grown by a minimum unit. A candidate becomes a verified meaning unit (MU2) if the teacher’s translation of the new source, given the committed text, is in the target language and reaches chrF ≥ 50 against the newly

aligned span of v . Its translation is due $\delta_t \geq \delta_s$ chunks after its source’s last chunk; a turn end commits the rest of the turn, so MODEL must detect it. Relative to the teacher’s alignment, targets thus never anticipate unheard speech, whereas sentence-level targets (Ma et al., 2019) require anticipating clause-final German and Korean verbs; but commit points depend on the whole turn, so MODEL imitates a privileged teacher. In a text-level Ko $\leftrightarrow$ En pilot (silver references; utterances much longer than CST-Bench-TAXI turns), MU2 roughly halves the lag of utterance-end commits (a LAAL variant), losing at most 2.2 chrF (Appendix G). *[Draft: planned: MU2 targets for En $\leftrightarrow$ De and at training scale; the German minimum unit and the δ_t values are set on held-out data]*

A token written after chunk k is time-stamped $(k+1)\Delta + \ell_{\text{enc}} + c$, so on schedule its latency after evidence ending at t is $q + \delta\Delta + \ell_{\text{enc}} + c$: quantisation $q \in (0, \Delta]$, stream delay $\delta \in \{\delta_s, \delta_t\}$, encoder look-ahead ℓ_{enc} (80 ms) and compute time c (zero on the ideal clock; Table 15). As turn ends force commits, EndOffset then stays near $q + \delta_t\Delta + \ell_{\text{enc}} + c$.

5.4. Training without real cross-lingual conversations

After single-speaker English and Korean ASR training (Stages 0–1), Stage 2 trains M1 for speaker-attributed transcription on simulated sessions with gaps and overlaps drawn more broadly than the benchmark’s; Stage 3 adds translations committed at turn ends (M2) or at MU2 (M3) (Table 17). Training excludes CST-Bench-TAXI and CST-Bench-Syn dialogues and voices (Appendix F). *[Draft: planned: Stages 2–3; German training data has yet to be obtained]*

6. Experiments

We answer RQ1–RQ4 (Section 1) and do not interpret differences inside the 95% CI (Section 4). *[Draft: planned: apart from the preliminary offline rows of Table 3 and ASR WERs of Table 5, no experiment of this section has been run]*

6.1. Setup

Systems. Offline systems translate each gold turn after it ends, without dialogue history (Table 12 and Appendix D): Qwen3.8-27B (“LLM”) translates the gold transcript (Gold $\rightarrow$ LLM) or that of Qwen3-ASR-1.7B (ASR $\rightarrow$ LLM; Shi et al., 2026), and Whisper-large-v3 translates into English (Whisper ST; Radford et al., 2023). Gold $\rightarrow$ LLM (MU2) streams gold transcripts to the LLM, which commits at the verified meaning units of Section 5.3, chosen non-causally with the whole turn in view. The consecutive cascade runs ASR $\rightarrow$ LLM on segments closed by its own VAD. Streaming systems are SeamlessStreaming (Seamless Communication et al., 2023); a streaming

cascade, whose LLM re-translates the growing transcript of the Nemotron 3.5 streaming RNN-T (NVIDIA, 2026) and commits the prefix shared by two consecutive outputs (LocalAgreement-2, LA-2; Liu et al., 2020; Polák et al., 2022); SeamlessM4T v2 (SEAMLESS Communication Team et al., 2025) with AlignAtt (M4T v2 + AlignAtt; Papi et al., 2023b); and, outside Table 3, StreamSpeech (De $\rightarrow$ En; Zhang et al., 2024) and, optionally, Hibiki-Zero (De $\rightarrow$ En, text output; Labiausse et al., 2026) and InfiniSST (En $\rightarrow$ De; Ouyang et al., 2025b). $2\times$ SeamlessStreaming, a naive bidirectional baseline, runs one instance per target language on the whole mix, without routing. MODEL runs as M2 (turn-end commits) and M3 (MU2 commits; Section 5.4), the latter at a low and a high δ_t . *[Draft: optional runs: Hibiki-Zero (public weights; only its text output is scored), InfiniSST only if its weights are released, and Canary-1B-v2 (Sekoyan et al., 2025) with AlignAtt]*

Wrappers and operating points. Single-direction systems run as one instance per target language behind a wrapper, forming a pipeline (Appendix D): the oracle wrapper cuts gold turns from the mix and routes each by its gold language; the realistic wrapper segments the stream with Silero VAD (Silero Team, 2024) and routes each segment by spoken language identification restricted to the pair. Baselines are not fine-tuned. Each streaming system sweeps its latency knob, which traces a quality–latency curve (Figure 4); tables report two operating points defined by the lag relative to turn length, $\rho = \text{StreamLAAL}/\bar{D}$, with $\bar{D}$ the mean source-turn duration of the direction: a *simultaneous* regime ($\rho \leq 0.5$, the main point; on CST-Bench-TAXI 1.54 s for De $\rightarrow$ En and 2.31 s for En $\rightarrow$ De) and a *consecutive* regime ($\rho \leq 1$, the lag of translating after each turn). Settings are fixed on the CST-Bench-Syn calibration split, before the test runs (Appendix D). *[Draft: until CST-Bench-Syn exists, documented defaults stand in for calibrated settings]*

Ranking and the oracle gap. We rank systems by COMET (Kocmi et al., 2021); on CST-Bench-TAXI, whose references lack case and punctuation, chrF replaces it if their rankings disagree (Appendix C). Latencies use the ideal clock; Table 15 (Appendix E) adds computation-aware latency, also for the ablation controls. A pipeline’s oracle gap $G = Q_{\text{oracle}} - Q_{\text{real}}$ is its COMET difference between the two wrappers. We decompose G for every main pipeline (consecutive cascade, SeamlessStreaming, streaming cascade) on both CST-Bench-TAXI and CST-Bench-Syn, so no pipeline is selected on test data; MODEL’s recovery $R = (Q_{\text{ours}} - Q_{\text{real}})/G$ is reported against the pipeline with the highest realistic COMET on each set (all pipelines in Appendix E), with a bootstrap CI only where the CI of G excludes 0.

Table 3. Real dialogues: CST-Bench-TAXI, L0 natural, ideal clock; systems of Section 6.1 at their main operating point (simultaneous budget, $\rho \leq 0.5$). Seg.: oracle = gold turns and languages; realistic = VAD and language-ID routing; none = the system segments the stream. Latencies in seconds (StreamLAAL: mean; others: median). 95% CIs (except BLEU) and single-direction streaming systems: Table 13. Blank: pending; -: not applicable. [†]Preliminary, scored by gold turn ID without resegmentation (Appendix C). [‡]0 by construction. [§]Non-causal.

System	Seg.	En→De					De→En					Both directions		
		COMET	BLEU	chrF	Stream-LAAL	End-Offset	COMET	BLEU	chrF	Stream-LAAL	End-Offset	Wrong dir. (%)	Switch latency	Empty turns (%)
Gold→LLM	oracle		48.31 [†]	71.72 [†]	4.62 [†]	0 [‡]		30.60 [†]	58.05 [†]	3.09 [†]	0 [‡]			
ASR→LLM	oracle		42.66 [†]	67.97 [†]	4.62 [†]	0 [‡]		28.11 [†]	56.61 [†]	3.09 [†]	0 [‡]			
Whisper ST	oracle	—	—	—	—	—		30.69 [†]	56.67 [†]	3.09 [†]	0 [‡]	—	—	—
Gold→LLM (MU2) [§]	oracle													
Consecutive cascade	realistic													
SeamlessStreaming	oracle													
	realistic													
2×SeamlessStreaming	none													
Streaming cascade	oracle													
	realistic													
M4T v2 + AlignAtt	oracle													
	realistic													
MODEL M2	none													
MODEL M3, low δ_t	none													
MODEL M3, high δ_t	none													

6.2. Real dialogues (RQ1, RQ3)

Table 3 and Figure 4 compare systems on CST-Bench-TAXI under L0 natural, the turn-taking that simultaneous translation should enable. Given oracle turns, Gold→LLM reaches 48.31[†] (En→De) and 30.60[†] (De→En) BLEU, ASR transcripts cost 5.65[†] and 2.49[†] of it, in line with a higher English than German ASR WER (Section 6.4), and Whisper ST reaches 30.69[†] into English. These offline systems look instantaneous on EndOffset, at most about 0.2[†] s even with measured, batched compute (Table 15), only because they receive oracle endpoints. The consecutive cascade, which must detect endpoints, makes listeners wait a median of [TBD] s at [TBD] COMET. Realistic segmentation costs the streaming systems [TBD] COMET (arrows in Figure 4), and MODEL reaches [TBD] COMET at a median EndOffset of [TBD] s without oracle input ($R = [TBD]\%$).

6.3. Where pipelines fail (RQ1, RQ2)

Table 4 replaces the oracle components of each main pipeline singly and jointly; a factor’s effect averages the COMET lost by replacing its component over both settings of the other factor, so the effects of segmentation and routing sum to G (Appendix E). We expect silence-based segmentation to merge turns across speaker changes, as a 500 ms threshold captures only 18–30% of between-speaker intervals in conversation (Heldner & Edlund, 2010) and CST-Bench-TAXI L0 natural draws a median gap of 0.223 s (Table 2). Under L0 natural, turn segmentation costs [TBD]–[TBD] COMET across pipelines and routing [TBD]–[TBD], at [TBD]% language-ID routing accuracy;

Table 4. Where the conversational loss comes from, for every main pipeline at its main operating point: oracle gap G (COMET with oracle minus realistic wrapper) and its parts from the oracle-component factorial, namely turn segmentation E_T and routing E_L under L0 natural, and the gain from two-channel separation E_S under L1 (TAXI: CST-Bench-TAXI; Syn: CST-Bench-Syn). COMET points, mean of both directions, ideal clock; definitions, CIs and all rows: Table 14 (Appendix E).

Pipeline	TAXI L0 nat.			TAXI L1		Syn En↔De L1	
	G	E_T	E_L	G	E_S	G	E_S
Consecutive cascade							
SeamlessStreaming							
Streaming cascade							

under L1, realistic segmentation yields [TBD]% wrong-direction words, and oracle separation recovers [TBD] COMET (failure types: Appendix E).

Timing conditions and language pairs. Appendix E contrasts the best realistic pipeline of each timing condition and language pair, under both wrappers, with MODEL (Figure 8) and over fixed gaps from −0.6 to 2 s (Figure 9). On CST-Bench-TAXI, the oracle gap is [TBD] COMET under L0 mediated, [TBD] under L0 natural and [TBD] under L1, where MODEL recovers [TBD]%; on CST-Bench-Syn L0 natural, it is [TBD] for Ko↔En against [TBD] for En↔De. The unit pilot accepts far fewer commit points for En→Ko than for Ko→En (21% vs. 54%; Appendix G), which verbal-final Korean or a chrF threshold stricter for Korean targets may explain.

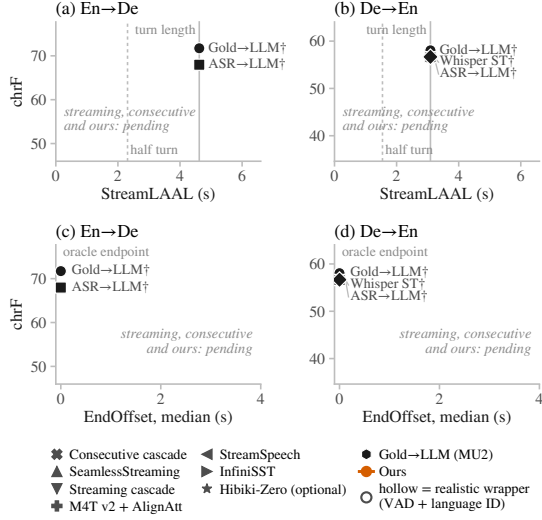


Figure 4. Quality (chrF until COMET is scored) vs. StreamLAAL (a, b) and median EndOffset (c, d) on CST-Bench-TAXI L0 natural. Black: offline systems and Gold→LLM (MU2). Pending: each baseline’s knob sweep with the oracle (filled) and the realistic (hollow) wrapper, and MODEL over δ_t (vermillion). Dashed and solid verticals: the simultaneous ($\rho = 0.5$, half a turn) and consecutive ($\rho = 1$, one turn) budgets. [†]Preliminary (Table 3).

6.4. Who said what: transcripts and speakers

Table 5 compares MODEL’s transcripts and speaker attribution with Qwen3-ASR-1.7B on oracle turns and with offline and streaming diarization cascades (Appendix D) built on pyannote (Bredin, 2023) and Streaming Sortformer (Park et al., 2025; Medennikov et al., 2025). MODEL reaches [TBD]% cpWER and [TBD]% DER on CST-Bench-TAXI L1 at $\delta_s = 2$. As each speaker keeps one language, the language reveals the speaker, so DER mostly measures boundary timing under short gaps and overlap.

6.5. Ablations: does the unified design help?

Table 6 changes one design choice of MODEL (M3) per row, starting from the same M1 checkpoint (Section 5.4; rows: Appendix E). Three controls keep the factorisation of Section 3. The *decoupled twin* fixes M1’s speaker tags as hard turn and direction decisions and then translates each segment in a separate pass of the same 1.7B thinker with the same commit policy, so translation cannot revise segmentation or routing. The *same-backbone cascade* (M1→LLM) translates M1 transcripts with the LLM under LA-2. Unlike the baselines, both share MODEL’s training on simulated sessions through M1, so we fixed them, before the test runs, as the tests of coupling turn-taking and translation. A size-matched cascade (ASR→1.7B LLM) separates design from scale by pairing Qwen3-ASR-1.7B with an LLM of about 1.7B parameters under LA-2. At matched StreamLAAL, MODEL differs in COMET under L0 natural and

Table 5. Who said what: error rates (%); systems: Section 6.4. L0: L0 natural; Syn: CST-Bench-Syn Ko↔En. DER: 0.25 s collar. Delay: median delay of transcript words. -: not applicable. [†]Preliminary.

Set	Metric	MODEL			
		Oracle ASR	Offline diar.	Streaming diar.	$\delta_s=2$ $\delta_s=4$
TAXI L0	WER En	13.78 [†]			
	WER De	7.33 [†]			
	cpWER				
	DER	—			
TAXI L1	Delay (ms)	—	—		
	cpWER				
Syn L1	DER	—			
	cpCER				

Table 6. Ablations of MODEL and controls (Section 6.5) on CST-Bench-TAXI, ideal clock; CIs omitted (comparisons with the controls carry CIs in the text). COMET: En→De/De→En; StreamLAAL: mean; EndOffset, switch latency: medians (s); wrong direction: %. [◦]Optional.

Variant	L0 natural		L1 turn-end overlap	
	COMET	Stream-LAAL	End-Offset	Wrong Switch dir. latency
MODEL (M3)				
turn-end commits				
no speaker tags				
no L1-like sessions [◦]				
fixed- δ_t models [◦]				
decoupled twin				
M1→LLM				
ASR→1.7B LLM				

[Draft: rows of runs that are cut will be removed, not left blank]

L1 by [TBD] and [TBD] from the twin, and by [TBD] and [TBD] from the same-backbone cascade. [Draft: planned: the decoupled twin is a new run (Appendix D)]

6.6. Validity checks (RQ4)

Appendix C tests rank agreement, as Kendall’s τ (tie-corrected, τ_b), between CST-Bench-TAXI and CST-Bench-Syn En↔De under L0 natural and across TTS editions, references, render seeds and metrics (Table 11 and Figure 7). Synthetic and real rankings give $\tau = [TBD]$, which we report whatever its value.

7. Limitations and Conclusion

Limitations. CST-Bench-TAXI has real speech but synthetic timing and overlap, for want of known public two-language recordings with natural overlap and references; it is small, single-domain, narrowband and close-talk, and

ties role to language (Sections 2 and 4). CST-Bench-Syn has fluent TTS speech, translationese on one side and silver Korean references, so it supports only relative conclusions, checked against CST-Bench-TAXI (Table 11).*[Draft: planned; CST-Bench-Syn and the turn-end overlap renders are not built yet]* Interruptions, barge-in and backchannels are left for future work. MODEL learns from simulated sessions and a privileged teacher (Section 5.3); only the decoupled twin and the same-backbone cascade test whether coupling turn-taking and translation helps (Section 6.5).*[Draft: planned; only the streaming ASR backbone is built, and all MODEL results are pending]*

Conclusion. CST requires deciding, from one mono stream, who speaks, which way to translate and when to commit: on CST-Bench-TAXI, segmentation and routing account for [TBD] and [TBD] COMET points of the best open pipeline’s oracle gap (RQ1), which moves from [TBD] to [TBD] between mediated gaps and turn-end overlap (RQ2), and MODEL recovers [TBD]% of it without oracle input at a median EndOffset of [TBD]s (RQ3). Synthetic and real rankings give $\tau = \text{[TBD]}$ (RQ4); real overlapped recordings and more language pairs come next.

Impact Statement

CST systems could let people who share no language talk through one device, and speaker-attributed transcripts could help deaf and hard-of-hearing people follow bilingual conversations. A mistranslated or misattributed turn can mislead in medical or legal settings, so deployments there should show the source transcript beside each translation and keep human interpreters available. Transcribing and attributing speech from one microphone could also enable covert monitoring; MODEL labels speakers only by arrival order, without identifying them, and any deployment should require the consent of everyone recorded. We redistribute no TAXI audio or text (only build scripts, configurations, checksums and per-turn scores), and all our text examples are invented except one KsponSpeech (Bang et al., 2020) transcript in Figure 11 (Appendix F).*[Draft: check that the KsponSpeech terms allow quoting this transcript; otherwise use an invented utterance there]* CST-Bench is for evaluation only; its synthetic voices are disclosed and clone no real person’s voice without consent, and its XDailyDialog-based subset is non-commercial.*[Draft: planned; CST-Bench-Syn is not built yet]* Outputs of the non-commercially licensed EXAONE-4.0 (LG AI Research, 2025) serve only the agreement check of Korean CST-Bench-Syn text and the unit pilot (Appendices B and G) and are neither released nor trained on; MODEL combines components licensed under OpenMDW-1.1 and Apache-2.0 (Table 20 and Appendix H).

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A. TAXI Datasheet and Why TAXI

This appendix documents CST-Bench-TAXI, the real track of CST-Bench (Section 4), and explains why it is built from TAXI. Table 7 lists its statistics, Figure 5 shows their distributions and Figure 6 shows one rendered session. Because TAXI may not be redistributed, we report aggregates only: no TAXI audio, transcript or translation appears in this paper, its figures or the released files.

Table 7. Full statistics of the CST-Bench-TAXI build (cstbench v0.1.0). Rendered turns are the 16 kHz turns after silence trimming, 0.1 s margins included; they are identical in L0 natural and L0 mediated. Untrimmed values refer to the usable source turn files. Turn files exclude pre-dialogue and hang-up items. Same-speaker successions: 9 arise because unusable turns between them were skipped, 5 are adjacent in the source. Reference words are those of the human translations of the column’s turns. Marks (SpeechDat notation): * hardly recognisable, ~ truncated, ** unintelligible. Percentiles use linear interpolation; the scorer’s p90 of m values instead takes the sorted value at the 0-based index $\min(m - 1, \lfloor 0.9m \rfloor)$ (Appendix C). –: not applicable or not reported; blank: pending.

Statistic	German (dispatcher)	English (client)	All
<i>Sessions and turns</i>			
Sessions	–	–	86
Turn files	374	323	697
Usable turns	349	291	640 (91.82%)
Unusable turns (skipped when rendering)	25	32	57
Sessions with ≥ 1 unusable turn	–	–	40
Usable turns per session: median (mean; range)	4 (4.06; 1–9)	3 (3.38; 0–8)	7 (7.44; 1–17)
Sessions opened by this speaker (first usable turn)	85	1	–
Direction switches: total (per session: median; range)	–	–	540 (6; 0–14)
Same-speaker successions (skipped / adjacent)	–	–	14 (9 / 5)
<i>Audio</i>			
Usable turn audio before trimming (min)	24.563	28.068	52.631
Speech after trimming (min)	17.953	22.425	40.378
Untrimmed turn (s): mean / median / p90	4.223 / 3.64 / 7.02	5.787 / 4.76 / 10.94	4.934 / 4.0 / 9.064
Rendered turn (s): mean / median	3.086 / 2.42	4.624 / 3.38	3.785 / 2.86
Rendered turn (s): p90 / max	5.492 / 12.64	9.92 / 17.98	7.76 / 17.98
Rendered turns longer than 5 s	–	–	144 (22.5%)
Rendered turns longer than 10 s	–	–	29 (4.5%)
Silence removed per turn (s): median leading / trailing	0.5 / 0.46	0.56 / 0.44	0.54 / 0.44
Speaking rate (words/s): trimmed (untrimmed) audio	2.997 (2.19)	2.666 (2.13)	–
<i>Text</i>			
Source words	3,228	3,587	6,815
Source words per turn: mean / median / p90 / max	9.25 / 8 / 17 / 34	12.33 / 10 / 25 / 37	–
Reference words	3,339	3,278	6,617
Reference words per turn: mean / median / max	9.57 / 8 / 44	11.26 / 10 / 38	–
Vocabulary, source / reference (lowercased types)	472 / 381	534 / 602	–
Filled-pause marks [fil]	60	172	–
Noise labels (NOI tier)	235	228	–
Other marks: * / ~ / **	53 / 8 / 1	61 / 4 / 11	–
Turns containing markup	101	135	–
<i>Rendered sessions, L0 natural / L0 mediated (gaps as drawn, between margin-inclusive turns)</i>			
Session audio (h)	–	–	0.737 / 0.998
Median session (s)	–	–	29.023 / 39.316
Speaker-change gap (s): mean / median, natural	–	–	0.257 / 0.223
Speaker-change gap (s): mean / median, mediated	–	–	1.995 / 2.026
Natural gaps at the 0.05 s floor	–	–	129 (23.89%)
Speaker-change silence between speech (s): median	–	–	–
Speech occupancy (%)	–	–	91.27 / 67.42
Overlap ratio	–	–	0.0 / 0.0

Source and usable turns. BAS TAXI 2.5 (Bavarian Archive for Speech Signals (BAS), 2001; Reichel et al., 2016), distributed through the BAS CLARIN repository, holds telephone dialogues between a German-speaking taxi dispatcher and an English-speaking client, each speaking their own language. Every turn is a separate 8 kHz, 16-bit mono file. Speakers took turns with a push-to-talk button, which prevented overlap and kept no inter-turn timing, so only the order

of the turns is known. Turns that passed the corpus validation carry a BAS Partitur annotation with orthographic words in SpeechDat notation (ORT tier), pronunciations (KAN), noise labels (NOI) and a human translation into the other language (TLN); turns rated as garbage (overlapping speech, meta talk, empty, noise only or wrong language) have none. We drop pre-dialogue and hang-up items, which leaves 697 dispatcher and client turn files in 86 sessions (IDs SES0037–SES0130, not contiguous), and keep a turn if its cleaned transcript is non-empty and it has a human translation whose direction matches the speaker’s language. Cleaning removes bracketed markers such as [fil] and unintelligible tokens (**), strips the marks of hardly recognisable (*) and truncated (~) words, and keeps word fragments. This leaves 640 usable turns (91.82%; German 349, English 291), exactly the turns that the validation list shipped with the corpus marks usable; 40 sessions contain at least one unusable turn, which the renderer skips. The corpus README reports 645 usable turns from the original validation, and the corpus’s revalidation report counts 640, as we do; catalogue entries give 94 dialogues, the number recorded, but version 2.3 removed the eight that have no usable turn. The build identifies speakers only by role, so we report no speaker counts or demographics. *[Draft: add speaker counts, recording conditions and translation guidelines from the corpus documentation]*

Composition. German turns are shorter than English ones: after silence trimming, the median turn lasts 2.42 s in German and 3.38 s in English, and the median number of source words per turn is 8 and 10. The rendered turns amount to 40.378 min of speech after silence trimming (52.631 min of usable turn audio before trimming). Over both languages, the median turn lasts 2.86 s, 22.5% of the turns exceed 5 s and 4.5% exceed 10 s (Figure 10 breaks latency down by turn length). On trimmed audio, the speaking rate is 2.997 words/s in German and 2.666 words/s in English. The transcripts mark what synthetic speech lacks: 60 German and 172 English filled pauses and 8 and 4 truncated words, and the NOI tier holds 235 and 228 noise labels. The human translations are lowercase and unpunctuated, so the scorer normalises hypotheses to this style (Appendix C). Each direction has only about 3.3k reference words (3,339 English words for the German turns and 3,278 German words for the English turns), so every comparison uses session-clustered bootstrap intervals (Appendix H).

Rendering. The build resamples the usable turns from 8 to 16 kHz with a polyphase filter and keeps the telephone band; it applies no bandwidth extension, which matters when reading ASR error rates. It trims each turn to the span from the first to the last 20 ms frame whose RMS exceeds a threshold of 0.1 times the 95th-percentile frame RMS (but at least 10^{-4}), widened by a 0.1 s margin on each side. Trimming removes a median of 0.54 s of leading and 0.44 s of trailing silence per turn; the totals, 416.06 s and 319.14 s, account exactly for the difference between the untrimmed (3157.86 s) and the trimmed audio (2422.66 s). *[Draft: a median leading silence of 0.88 s (about 0.9 s in the code docstring) is not reproduced by this build; do not quote it]* Only the timing is sampled, as in simulated conversations for diarization training (Landini et al., 2022); the turns and their order are real. The renderer places the turns in dialogue order after a 0.5 s lead, separates consecutive turns of one speaker by a pause drawn from $\mathcal{U}[0.3, 0.8]$ s, appends a 0.5 s tail and sums the two speaker channels, clipped to $[-1, 1]$. Gaps and pauses separate the margin-inclusive turn intervals, so the silence between the energetic frames of two turns is 0.2 s longer than drawn unless a margin was cut at a file edge; the gap statistics below are the drawn values, and Table 7 also lists the silence between the energetic frames (speech) of two turns. At a speaker change, L0 natural draws the gap from $\mathcal{N}(0.2, 0.25^2)$ s clipped to $[0.05, 1.0]$ s, close to the typical gap of about 200 ms (Levinson & Torreira, 2015), which varies little across languages (Stivers et al., 2009); L0 mediated draws it from $\mathcal{U}[1, 3]$ s, for a listener who reads the translation before answering. At the 540 speaker changes, natural gaps have a mean of 0.257 s and a median of 0.223 s, and 129 of them (23.89%) sit at the 0.05 s floor, against about 27.4% expected from clipping the normal; mediated gaps have a mean of 1.995 s and a median of 2.026 s (Figure 5c). The renders last 0.737 h (natural; median session 29.023 s; speech occupancy 91.27%) and 0.998 h (mediated; 39.316 s; 67.42%), and the overlap ratio is 0.0 in all 86 sessions of both (Table 7). Real conversation overlaps at many speaker changes (Heldner & Edlund, 2010); L1 therefore starts the next turn $\mathcal{U}[0.2, 0.8]$ s before the previous one ends at each speaker change, under its own configuration name and checksums, and its overlap will be reported per turn and, once the turns are force-aligned (Shi et al., 2026), per word. *[Draft: planned; the L1 renderer is not built yet]*

Caveats. Speech, content and turn order are real, but every gap is synthetic and the corpus has no overlap; overlap claims therefore rest on the L1 renders and on CST-Bench-Syn (Appendix B). CST-Bench-TAXI stands in for the face-to-face case: in the original calls the two sides arrived on separate channels, the situation of the oracle-separation diagnostic, so the mono mix models one shared device rather than the calls, and its close-talk, narrowband speech lacks the room acoustics and level differences of such a device. Turn intervals include the 0.1 s margins, so the turn boundaries to which EndOffset and switch latency refer can be off by up to about 0.1 s, and energy trimming may clip soft onsets; forced-aligned word times will check these boundaries. *[Draft: planned; the TAXI turns are not force-aligned yet]* Language and role are confounded

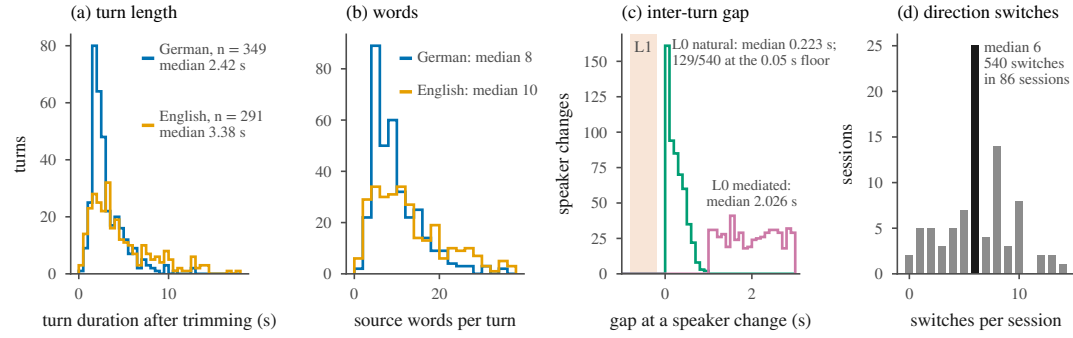


Figure 5. CST-Bench-TAXI statistics (86 sessions, 640 usable turns; aggregates only, as TAXI may not be redistributed). (a) Turn duration after silence trimming (0.1 s margins included), by source language. (b) Source words per turn. (c) Gaps drawn at the 540 speaker changes under L0 natural and L0 mediated; the shaded band of negative gaps marks where the planned L1 condition starts the next turn, 0.2–0.8 s before the previous turn ends. (d) Direction switches per session; the median (6) is drawn in black, and two sessions have no switch.

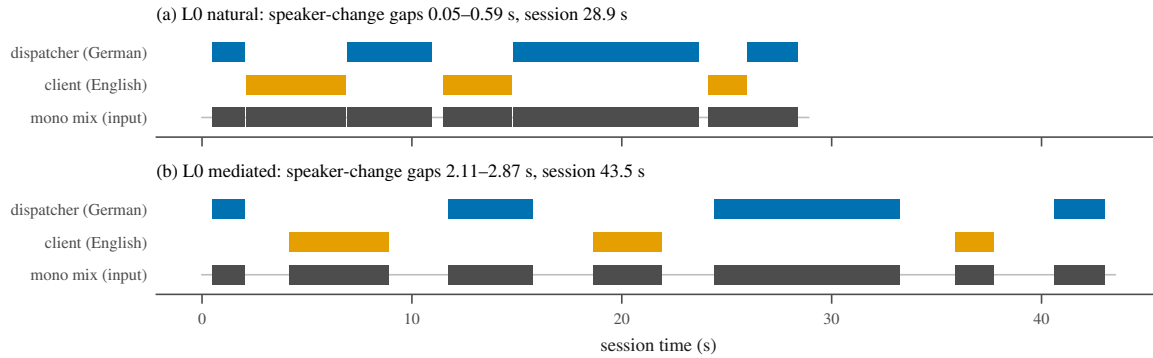


Figure 6. Session SES0104 (7 turns) rendered under (a) L0 natural (speaker-change gaps 0.05–0.59 s; session 28.9 s, close to the median session of 29.023 s) and (b) L0 mediated (gaps 2.11–2.87 s; session 43.5 s). Lanes show the German dispatcher, the English client and the mono mix, which is all a system receives. Speech and turn order are real; the timing is synthetic. Only turn intervals are shown, no audio or text.

(German is always the dispatcher), and the dispatcher opens 85 of the 86 sessions, a trivial prior for the first direction. The realistic wrappers therefore route by spoken-language identification (Appendix D); as the language also reveals the direction, CST-Bench-TAXI tests the timing of speaker changes more than their detection, and CST-Bench-Syn renders both speaker–language assignments. Sessions SES0085 and SES0095 hold a single usable turn, in German; they have no direction switch and are excluded from switch metrics. Fourteen turns follow a turn of the same speaker, 9 because unusable turns in between were skipped and 5 because the two turns are adjacent in the source; a pause separates them, and they do not count as switches. Finally, TAXI serves only for evaluation: no threshold, operating point or model is tuned on it.

Release and reproducibility. TAXI is free of charge under a scientific licence, and a commercial licence is available; redistribution, even in part, requires the written permission of the copyright holders. We therefore release session and turn IDs, configurations, build scripts, checksums and per-turn scores keyed by turn ID, but no audio, text or system output derived from TAXI (Table 20). Each rendered session directory holds `mix.wav` (the single-channel system input), `2ch.wav` (one channel per speaker, for analysis and oracle diagnostics only) and `timeline.json` (turn intervals with margins, gaps, removed silence, transcripts and references). The channel order of `2ch.wav` follows the `channels` field of the timeline: channel 0 is the English client and channel 1 the German dispatcher.[\[Draft: the package README states the reverse order; fix it before release\]](#) Per-session seeds derived from the session IDs and Python’s `random` module make the timelines identical on every platform (Section 4). `cstbench verify` requires the SHA-256 digest of every timeline (as sorted-key JSON) to match exactly and reports audio-hash differences separately, because audio hashes can depend on the `numpy`, `scipy` and `libsndfile` versions (the reference hashes were made with `numpy` 1.24.3, `scipy` 1.10.1 and `libsndfile` 1.2.2). A local rebuild matched 86/86 timeline digests and 86/86 `mix.wav` and `2ch.wav` hashes in both configurations. Released configurations are frozen, and any change gets a new name. With the corpus in `TAXI_DIR` (one folder per session), the track is built and verified by

```
cstbench build-taxi --root TAXI_DIR --out OUT_DIR
cstbench verify --out OUT_DIR/taxi-natural --reference checksums/taxi-natural.json
cstbench verify --out OUT_DIR/taxi-mediated --reference checksums/taxi-mediated.json
```

Why TAXI. A real CST test set needs one conversation between at least two real speakers who use different languages, with shared multi-turn context, per-turn audio, the identity and language of each speaker, turn order and transcripts; reference translations, turn timing, session audio and per-speaker channels are desirable. Parallel speech read from scripts, separately recorded sides of a text dialogue, merged monolingual corpora and utterance sets whose order or context cannot be recovered do not qualify. Table 8 checks the candidates against these requirements. MSLT (Federmann & Lewis, 2016; 2017) was recorded from bilingual calls, but the English side of each call was discarded. DRAL (Ward et al., 2023; 2024; Avila & Ward, 2023) and mDRAL (Seamless Communication et al., 2023) pair fragments of one-language conversations with re-enactments in the other language. InCroMin (Čechovič et al., 2025) records real meetings between people without a common language, but they talk through machine speech translation, each participant is recorded on a separate track, and no English–German pair is included. CS-Dialogue (Zhou et al., 2025) holds about 104 h of Mandarin–English code-switching without translations. The Verbmobil II interpreter volumes (Bavarian Archive for Speech Signals (BAS), 2000), from the Verbmobil project (Wahlster, 2000), contain German–English and German–Japanese dialogues and a human-interpreter subset, but their speech is distributed on paid media orders (the repository edition that is free for research holds the annotations)[\[Draft: verify the VM2 audio access terms with BAS\]](#), and the Fisher–CALLHOME Spanish–English corpus (Post et al., 2013) has a single source language. Among these candidates, only TAXI combines real two-language dialogues, human translations of both sides and speech that is free to download for research. What it lacks, turn timing and overlap, is exactly what the renderer controls.

Table 8. Why CST-Bench-TAXI: real bilingual conversational speech resources checked against the requirements of a real CST test set (citations in the text of Appendix A). ✓ yes, ◦ partly, ✗ no; –: not assessed or not applicable.

Corpus	Same conversation, two languages	Real spontaneous speech	Human translations	Per-turn audio / timing	Overlap	Access / licence	Verdict
BAS TAXI 2.5	✓ German dispatcher, English client	✓	✓ both sides (TLN tier)	◦ per-turn files; no timing (push-to-talk)	✗ none	free scientific licence; no redistribution	used (real track)
MSLT	◦ bilingual calls; English side discarded	✓	◦ released side only	one side only	–	–	unsuitable (one-sided release)
DRAL / mDRAL	✗ one language per conversation; fragments re-enacted in the other	◦ originals only	◦ spoken re-enactments	paired fragments	–	LDC (DRAL)	unsuitable (re-enacted)
InCroMin	✓ meetings mediated by machine speech translation	✓	◦ corrected machine translations into English	per-participant tracks	–	CC BY 4.0	related work (12 languages, no English–German pair)
CS-Dialogue	◦ Mandarin–English code-switching within speakers	✓	✗	full dialogues	–	–	unsuitable (about 104 h, no translations)
Verbmobil II interpreter volumes	✓ German–English and German–Japanese; human-interpreter subset	✓	–	–	–	speech on paid media	not used
Fisher–CALLHOME Spanish–English ST	✗ one source language (Spanish)	✓	✓ crowd-sourced, into English	–	–	LDC	related (one source language)

B. CST-Bench-Syn Construction and Data Card

[Draft: planned: CST-Bench-Syn is not built yet; this appendix specifies its construction, and every measured value in Table 9 is pending]

CST-Bench-Syn is an evaluation-only set of synthetic (TTS) speech. It adds what CST-Bench-TAXI cannot provide: turn-end overlap that can be measured at the word level, a distant language pair, and both speaker–language assignments, which remove TAXI’s first-speaker prior and its tie between language and role. It is not real conversation, does not meet the requirements for a real test set (Appendix A) and is translationese on one side of every pair. We therefore draw only relative conclusions from it (gaps between protocols and system rankings, never absolute scores across directions or pairs) and test whether it ranks systems as CST-Bench-TAXI does (Table 11 and Figure 7). There is no training or development split: a calibration split of about 5% of the sessions, disjoint from the evaluation sessions, is the only data on which wrapper thresholds and operating points are chosen, with one rule for all systems. Table 9 is the data card.

Table 9. CST-Bench-Syn data card (planned). The set is not built yet: blank cells will hold measured values, and the last column gives plan targets, which are not results. A session is one dialogue under one speaker–language assignment; every dialogue is rendered under both. The re-recognition error is the WER for English and German and the CER for Korean; turns above the target are re-synthesised. MT: machine translation; n/a: not applicable; –: no target.

Statistic	En↔De		Ko↔En		Plan target
	XDailyDialog	BConTrasT	XDailyDialog	BConTrasT	
<i>Size</i>					
Dialogues / sessions					–
Turns per direction					–
Speech per timing condition (h)					–
Calibration split (% of sessions)					about 5
<i>Speech</i>					
Voices per language and edition					–
Re-recognition error (%)					≤ 5
Re-synthesised turns (%)					–
Speaker cosine, same / other speaker					–
TTS word times vs. forced alignment					–
<i>Text</i>					
Reference type	human	De: post-edited MT En: human	Ko: consensus MT En: human	Ko: consensus MT En: human	–
Korean consensus pass rate (%)	n/a	n/a			≥ 85
Korean spot-check error rate (%)	n/a	n/a			–
Contamination-probe agreement					–
Licence	CC BY-NC-SA 4.0	CC BY-SA 4.0	CC BY-NC-SA 4.0	CC BY-SA 4.0	–

Pairs and speaker assignment. En↔De is the main pair: it shares its languages with CST-Bench-TAXI, which makes the sim-to-real comparison possible. Ko↔En adds a distant word order. Every dialogue is rendered under both speaker–language assignments (speaker *A* speaks the first language and *B* the second, and vice versa), and the same dialogues serve both pairs, so differences between pairs come from the languages rather than from the content. We exclude German–Korean, which almost no baseline supports, and use English as the pivot.

Dialogues. XDailyDialog (Liu et al., 2023) provides professional translations of the everyday dialogues of DailyDialog (Li et al., 2017), and BConTrasT (Farajian et al., 2020) provides English–German versions of task-oriented Taskmaster-1 dialogues (Byrne et al., 2019) between a customer and an agent in six domains, machine-translated into German and post-edited by native German speakers. Speaker *A*’s turns come from one language version of a dialogue and *B*’s from the other, so both share one context and the parallel version of each turn is its reference. We select dialogues of 6–16 turns and balance the domains, the turn lengths and the share of dialogues that contain numbers or named entities. XDailyDialog turns are short (about 11.5 tokens on average), so the selection favours multi-sentence turns, which leave room for turn-end overlap. No parallel corpus that cannot be redistributed is used.

Contamination. DailyDialog and Taskmaster-1 text may be in the pretraining data of the LLMs that serve as baselines and teachers. For each of them, a probe gives the opening turns of a selected dialogue, lets the model continue it and

measures the agreement of the continuation with the original (Table 9). Generation prompts, models and seeds are released. No DailyDialog, Taskmaster-1 or TAXI text and no CST-Bench-Syn dialogue or voice is used to train MODEL (Section 5).

References. English–German references are the parallel versions of each turn: the English side is original, and the German side is a professional translation (XDailyDialog) or a post-edited machine translation (BConTrasT). Neither source contains Korean, so every Korean text, both the Korean turns that are synthesised and the Korean references of English turns, is translated from English by three independent systems: Qwen3.8-27B, MADLAD-400-10B-MT (Kudugunta et al., 2023) and EXAONE-4.0 (LG AI Research, 2025). A text passes if at least two of the three translations agree, as judged by MetricX-24 in its reference-free mode (Juraska et al., 2024), CometKiwi (Rei et al., 2022b), the LaBSE similarity (Feng et al., 2022) of back-translations, and matching numbers and named entities; text that fails is removed, and we report the pass rate. The released Korean text is the MADLAD-400 translation, as no evaluated system belongs to that model family: Qwen3.8-27B is also a baseline translator and the teacher of our commit targets (Section 5), and EXAONE-4.0, whose licence restricts its outputs to non-commercial use, serves only in the agreement check, so its outputs are neither released nor used for training. *[Draft: fix the rule for text whose MADLAD-400 translation is the dissenting one]* Native Korean speakers annotate the errors in a fixed random sample with a light MQM-style scheme (Freitag et al., 2021). Ko↔En enters the main results only if at least 85% of the Korean text passes and the speech meets the re-recognition target below; otherwise it is reported as a pilot. For En↔De, silver references built in the same way test whether gold and silver references rank systems alike (Table 11). Gold and silver segments are aggregated separately, and reference-free CometKiwi is reported next to the reference-based metrics.

Speech synthesis and quality assurance. A commercial edition carries the main results: Azure neural TTS (Microsoft, 2026) with native preset voices for en-US, de-DE and ko-KR, balanced by gender, and word boundaries from the synthesiser. An open edition, CosyVoice 3 (Du et al., 2025) or Qwen3-TTS (Hu et al., 2026) (both with Apache-2.0 weights), covers a subset and tests whether the TTS engine changes system rankings (Table 11). A pilot of 50 sentences per language admits an engine only if its re-recognition error is at most 5%. Each speaker keeps one voice throughout a dialogue. Voices are presets, designed voices or voices cloned from CC0 reference recordings; we clone no real person’s voice without consent and no voice across languages. Every synthesised turn is re-recognised with Whisper-large-v3 (Radford et al., 2023) and Qwen3-ASR (Shi et al., 2026); turns with a WER (CER for Korean) above 5% are re-synthesised, and we report the error and the re-synthesis rate. Speaker consistency is measured by the cosine similarity of speaker embeddings within and between speakers. The synthesiser’s word boundaries are compared with forced alignment (Qwen3-ForcedAligner; Shi et al., 2026), and we report the distribution of their differences, because word-level overlap and the reference points of latency depend on these times.

Rendering and release. Sessions are rendered with the same renderer and rules as CST-Bench-TAXI (Section 4) under L0 natural and L1, optionally also under L0 mediated, each with its own configuration name and checksums; overlap is measured on turn intervals and on word boundaries. Mixes are clean: noise and reverberation variants are outside this release, and a telephone-band variant is added only if CST-Bench-Syn and CST-Bench-TAXI rank systems differently. Each session comes with the mono mix, per-speaker channels (for analysis and oracle diagnostics only), a timeline with turn and word times, and references flagged as gold or silver. A data card discloses that the speech is synthetic and lists the sources, licences, generation models, prompts, seeds and QA results. CC BY-NC-SA 4.0 and CC BY-SA 4.0 are incompatible ShareAlike licences, so the XDailyDialog-based and the BConTrasT-based sessions form two subsets, each under the licence of its source (Taskmaster-1, on which BConTrasT builds, is CC BY 4.0). The commercial edition is synthesised only on a paid tier (Table 20). Instead of a fixed size in hours, CST-Bench-Syn is sized by a power estimate from the variance of TAXI segment scores for the main comparisons, fixed before the test runs: a first tranche is built and extended only if its confidence intervals are too wide.

Out of scope. This release leaves out the parts of an earlier design that targeted interruptions and barge-in: episodes of two to four history turns, an interruption event and one or two resolution turns; information-causal event generation with three difficulty levels; sizing by at least 300 events per combination of event type, difficulty and pair, which implied about 10–12 h per pair; partial references for interrupted turns; leakage checks for generated events; and a high-overlap condition with at least 25% overlap. Interruptions remain future work (Section 7).

Limitations. TTS prosody lacks disfluencies, reactions to the partner and the Lombard effect; the dialogue text is largely planned rather than spontaneous; one side of every pair is translationese; the Korean references are silver; and wrapper

thresholds calibrated on clean 16 kHz TTS may suit telephone-band CST-Bench-TAXI less well. Rank agreement with CST-Bench-TAXI and between the two TTS editions (Table 11) indicates how far conclusions drawn on CST-Bench-Syn transfer to real speech.

C. Scorer Specification and Metric Validation

This appendix specifies `cstbench eval` precisely enough to reimplement it and describes how we test that its metrics measure what they are meant to measure. As in Sections 3 and 4, turn n of a session of duration T has speaker s_n , interval $[b_n, e_n]$, duration $L_n = e_n - b_n$ and a reference translation r_n of $|r_n|$ words. *[Draft: the released scorer implements the submission format, normalisation, resegmentation, oracle mode, BLEU, chrF, StreamLAAL, EndOffset, empty turns and wrong-language words given gold turn IDs; a pinned num2words, COMET, switch latency, the length ratio, the wrong-direction detector, the latency variants, the bootstrap and all transcript and speaker metrics are planned]*

C.1. Submission, clocks and normalisation

Submission. A system submits one JSON line per committed piece with four fields: session ID, output language, emission time t in seconds from the start of `mix.wav`, and text. The output language selects the direction (English output translates German speech), so no speaker or direction label is needed. Pieces are never retracted, and every line counts as committed. The scorer sorts the pieces of each session by emission time, keeping file order for ties, splits each normalised piece into words and gives every word the time of its piece. An optional gold turn ID per piece selects the oracle mode described below. Further fields are kept, and any of them can serve as the clock; pieces of unknown sessions are reported and ignored.

Clocks. On the ideal clock, t is the audio consumed when the piece is committed; the computation-aware clock (Ma et al., 2020c) adds processing time measured on fixed hardware (Table 15). Unlike SimulEval (Ma et al., 2020a), which runs a policy on each source segment, `cstbench eval` sees only committed pieces and their times, so they can come from any streaming harness, such as `simulstream` (Gaido et al., 2025), or from a unified model. On CST-Bench-TAXI, turn intervals include the 0.1 s margins of energy trimming (Appendix A), which can move the reference points of every latency metric by up to about 0.1 s.

Normalisation. Hypotheses and references are normalised alike, to the style of the TAXI references (lowercase, unpunctuated, numbers in words): (i) Unicode NFC; (ii) each number, i.e., a digit string with an optional decimal part, is spelled out with `num2words` in the output language, reading a decimal comma as a decimal point in German and dropping commas as thousands separators in the other languages; (iii) lowercasing; (iv) every character other than a letter, a digit, whitespace or a word-internal apostrophe becomes a space; (v) splitting at whitespace. The scorer depends on a pinned `num2words` version, because without the package step (ii) would be skipped silently. *[Draft: planned: pinning num2words; the released scorer lists it only as an optional dependency (version 0.5 or later) and skips step (ii) silently if it is missing]*

Quality. Per direction, BLEU (Papineni et al., 2002) and chrF (Popović, 2015) are sacreBLEU corpus scores (Post, 2018) over the turns of that direction, with default settings except for sacreBLEU’s Korean tokenizer for BLEU into Korean; we report the signatures and two decimals. An empty turn enters as an empty hypothesis and lowers both scores. COMET with the `wmt22-comet-da` model (Rei et al., 2020; 2022a) scores the same resegmented turns, with the gold transcript as the source. A per-turn file holds each turn’s source, hypothesis, reference and latencies; for CST-Bench-TAXI, only per-turn scores keyed by turn ID are released (Appendix H). *[Draft: planned: COMET inside the scorer and the Korean tokenizer; the released scorer leaves COMET to external tools reading the per-turn file]*

C.2. Resegmentation and oracle mode

For each session and output language λ , `cstbench eval` assigns the hypothesis words to the turns $\mathcal{N}_\lambda$ that are translated into λ , with the minimum-edit-distance criterion of `mwerSegmenter` (Matusov et al., 2005) as used in stream-level evaluation (Iranzo-Sánchez et al., 2021; Papi et al., 2024):

1. order the hypothesis words $w_1, \dots, w_M$ by emission time;
2. concatenate the normalised references r_n , $n \in \mathcal{N}_\lambda$, in dialogue order into $u_1, \dots, u_K$, and record the turn $o(k)$ of

each reference word u_k ;

3. align $w_{1:M}$ to $u_{1:K}$ by word-level Levenshtein alignment with unit costs; on ties, the backtrace prefers a match or substitution to an insertion, and an insertion to a deletion;
4. assign each w_i aligned to some u_k , as a match or a substitution, to turn $o(k)$; assign each inserted word to the turn of the preceding aligned word or, if it precedes all aligned words, to the turn of the first aligned word.

Turn n thus receives the words h_n , with emission times $\tau_{n,1} \leq \dots \leq \tau_{n,|h_n|}$, and is empty if h_n is empty. Resegmentation looks only at text, so a word can land in a turn that begins after the word was emitted, and latency on resegmented output inherits a segmentation-related bias (Polák et al., 2025).

Oracle mode. If every piece of a session and direction carries a gold turn ID, the scorer skips resegmentation and assigns words to their turns directly; words whose ID is not a turn of direction λ count as wrong-language words and are not scored. This mode is only a diagnostic (Section 3): scoring the same output both ways measures how much resegmentation alone moves the scores. Final results are resegmented, with any turn IDs removed from the hypotheses first; the preliminary rows of Table 3 were scored in oracle mode and will be re-scored.

C.3. Latency

For a non-empty turn n , let $d_{n,j} = \max(0, \tau_{n,j} - b_n)$ be the delay of its j -th word from the turn start, $\rho_n = L_n / \max(|h_n|, |r_n|)$ the rate, and J_n the first j with $d_{n,j} \geq L_n$, or $|h_n|$ if there is none. Then

$$\text{StreamLAAL}_n = \frac{1}{J_n} \sum_{j=1}^{J_n} (d_{n,j} - (j-1) \rho_n), \quad (6)$$

$$\text{EndOffset}_n = \tau_{n,|h_n|} - e_n, \quad (7)$$

$$\text{SwitchLatency}_n = d_{n,1} \quad \text{for } n > 1 \text{ with } s_n \neq s_{n-1}. \quad (8)$$

Equation (6) is LAAL (Papi et al., 2022), the length-adaptive form of Average Lagging (Ma et al., 2019), applied to resegmented turns, i.e., StreamLAAL (Papi et al., 2024). Two cases show what the metrics separate. If a turn’s translation is committed at e_n , every delay equals L_n , so $J_n = 1$, $\text{StreamLAAL}_n = L_n$ and $\text{EndOffset}_n = 0$: offline systems on oracle turns score the mean turn duration, 4.624 s for En→De and 3.086 s for De→En on CST-Bench-TAXI. If instead the j -th of the $|r_n|$ reference words is committed at $b_n + jL_n/|r_n|$, then $\text{StreamLAAL}_n = L_n/|r_n|$, the duration of one word, while EndOffset_n is still 0. StreamLAAL thus measures the lag within a turn and EndOffset the wait after it. EndOffset is negative if a turn’s output ends before the speaker does, through anticipation or truncation, so we always report it with quality and length ratio. Neither waiting metric is new: EndOffset is SimulEval’s EndOffset, the delay of the last output after the end of the source (Ma et al., 2020a), which SeamlessStreaming also reports as its ending offset (Seamless Communication et al., 2023), computed here per turn; switch latency (Equation (8)) is SimulEval’s StartOffset, the delay of the first output, taken only at turns that change direction. It is undefined for the first turn of a session, so the two single-turn CST-Bench-TAXI sessions contribute none; and as 540 of the 554 later CST-Bench-TAXI turns follow a speaker change, switch latency on CST-Bench-TAXI is close to the first-word delay of all later turns.

Aggregation. Per direction, we report the mean, median and 90th percentile over non-empty turns, to three decimals. The median averages the two middle values of an even count, and the 90th percentile of m values is the sorted value at the 0-based index $\min(m-1, \lfloor 0.9m \rfloor)$. The empty-turn rate is reported next to every latency.

Penalised variant. Averaging over non-empty turns lets a system improve its latency by dropping hard turns. The penalised variant therefore scores an empty turn as if its whole reference were committed at the session end: $\text{EndOffset}_n = T - e_n$ and $\text{StreamLAAL}_n = \text{SwitchLatency}_n = T - b_n$. On the ideal clock no piece comes later than T , so these are the worst values any output can obtain, and dropping a turn never lowers its latency.

Exactly matched words. Unrelated output can lower StreamLAAL (row 2 of Table 10); one mechanism is that a word assigned to a turn that begins after the word was emitted gets delay 0. We therefore also report Equations (6) and (7) with h_n restricted to the words that exactly match their aligned reference word, so that unrelated words cannot contribute. [Draft: planned: switch latency, the penalised variant and the matched-word variant]

C.4. Errors and auxiliary outputs

Empty turns and length ratio. A turn is empty if no word is assigned to it. Per direction, we report the share of empty turns and the ratio of hypothesis to reference words. *[Draft: planned: the length ratio; the released scorer reports only the numbers of turns and of empty turns]*

Wrong-direction words. In oracle mode, wrong-direction words are the wrong-language words defined above. Without turn IDs, a piece (t_i, λ_i, y_i) is checked against the turns open at its emission time, $\mathcal{A}(t_i) = \{n : b_n \leq t_i \leq e_n + W\}$, with a window W fixed before any system is scored. All words of the piece are wrong-direction words if (a) *wrong channel*: $\mathcal{A}(t_i)$ is non-empty and all its turns are spoken in λ_i , so no open turn needs a translation into λ_i ; (b) *wrong language*: a text language identifier restricted to the session’s two languages assigns y_i to the language other than λ_i ; or (c) *copy-through*: y_i is closer to the transcript of an open turn spoken in λ_i than to the reference of every open turn translated into λ_i , where the distance from y_i to a text is the smallest word edit distance between y_i and a span of that text, divided by the length of y_i . Under turn-end overlap, turns in both languages are open at once, so (a) cannot fire and (b) and (c) decide. Rows 2 and 8 of Table 10 test (b) and (c), and the delayed rows check that late but correct output is not flagged. We report wrong-direction words as a percentage of all committed words and compare the detector with the oracle-mode count wherever turn IDs exist. *[Draft: planned: the detector, its window W and the text language identifier; the released scorer counts wrong-language words only given gold turn IDs]*

Transcripts and speakers. For systems with a transcript channel, we report WER per language after the same normalisation; cpWER (Watanabe et al., 2020), which compares each speaker’s concatenated words under the best mapping between hypothesis and reference speakers, so that arrival-order labels need no fixed identity; and its time-constrained variant tcpWER from MeetEval (von Neumann et al., 2023). For Korean, cpCER applies cpWER to characters instead of words. DER uses collars of 0 and 0.25 s at 80 ms resolution. The main text reports WER, cpWER (cpCER on CST-Bench-Syn Ko $\leftrightarrow$ En) and DER (Table 5); tcpWER and DER without a collar are in Appendix E. *[Draft: planned]*

Speaker-attributed BLEU. DiariST scores translations per speaker after mapping hypothesis speakers to reference speakers (Yang et al., 2024). In CST each speaker uses one language, so the reference translations of speaker A are exactly those of direction $\ell_A \rightarrow \ell_B$. For a system that labels its translations with speakers, speaker-attributed BLEU therefore scores the same hypothesis–reference pairs as per-direction BLEU whenever every piece labelled A is in ℓ_B and every piece labelled B is in ℓ_A . The two differ only on pieces whose label contradicts their language; these are attribution or direction errors, which the wrong-direction rate and cpWER already count. We therefore compute speaker-attributed BLEU only as a consistency check for systems with speaker labels.

C.5. Metric meta-evaluation

We perturb the reference output on CST-Bench-TAXI L0 natural and check that BLEU, chrF, StreamLAAL, EndOffset and the wrong-direction count each move as expected while the others stay put (Table 10); the expected changes are fixed in advance. COMET and switch latency are not yet in the table, as the released scorer does not compute them. Row 1 commits every reference translation at its turn end. Row 2 emits each turn’s source transcript as its translation, as a system that passes speech through untranslated would. Rows 3–5 delay row 1 by 0.5, 1 and 2 s. Row 6 commits the j -th reference word of turn n at $b_n + jL_n/|r_n|$, and row 7 drops the last $\lfloor |r_n|/3 \rfloor$ words of every turn of row 6. Row 8 simulates a VAD that merges two turns across a speaker change, at a seeded subset of speaker changes: the first turn’s reference is committed at the end of the second turn, followed on the same channel by the second turn’s source transcript, as if the merged segment had been routed by the first turn’s language. Row 9 removes the output of a seeded 10% of turns. Row 10 rescores rows 1, 3–7 and 9, whose words all come from the right turns, with gold turn IDs and reports the largest change. No final system result is reported until every row behaves as expected or the deviation is explained; the preliminary system results have not yet passed this check and will be re-run. We repeat the comparison of row 10 for every system run under the oracle wrapper, whose pieces carry turn IDs.

Rows 1 and 2 exist so far only as preliminary runs ($\dagger$; to be re-run under the final protocol). Row 1 behaves as derived above: BLEU and chrF are 100.0 $\dagger$, EndOffset is 0.0 $\dagger$ s, no word counts as wrong, and the mean StreamLAAL equals the mean turn duration (4.624 $\dagger$ s for En $\rightarrow$ De, 3.086 $\dagger$ s for De $\rightarrow$ En). Row 2 drives BLEU down to 2.38 $\dagger$ for En $\rightarrow$ De and 0.39 $\dagger$ for De $\rightarrow$ En, while chrF keeps 27.34 $\dagger$ and 21.86 $\dagger$ from shared character n-grams. However, it *lowers* the mean StreamLAAL below that of the reference (4.366 $\dagger$ vs. 4.624 $\dagger$ s for En $\rightarrow$ De; 2.276 $\dagger$ vs. 3.086 $\dagger$ s for De $\rightarrow$ En) and raises the mean EndOffset

Table 10. Metric meta-evaluation on CST-Bench-TAXI, L0 natural: controlled perturbations of the reference translations, with the change expected for each metric fixed in advance (construction of each row in Appendix C). Q: BLEU and chrF; L: mean StreamLAAL (s); E: mean EndOffset (s), both over non-empty turns; W: number of wrong-direction words; $L_n, |r_n|$: duration and reference length of turn n ; =: unchanged; $\uparrow/\downarrow$: increases/decreases. † Preliminary; to be re-run under the final protocol. *The current scorer counts such words only given gold turn IDs. Blank cells are pending.

Perturbation	Expected change	En→De			De→En			W
		BLEU/chrF	L	E	BLEU/chrF	L	E	
1 Reference at turn end	Q = 100; L = mean L_n ; E = 0; W = 0	100.0/100.0 †	4.624 †	0.0 †	100.0/100.0 †	3.086 †	0.0 †	0 †
2 Source transcript passed through untranslated	Q $\approx$ 0; L, E not below row 1; W = all words	2.38/27.34 †	4.366 †	2.576 †	0.39/21.86 †	2.276 †	1.421 †	0 † *
3 Row 1 delayed by 0.5 s	Q =; L, E +0.5 s; W = 0							
4 Row 1 delayed by 1 s	Q =; L, E +1 s; W = 0							
5 Row 1 delayed by 2 s	Q =; L, E +2 s; W = 0							
6 Reference spread evenly over the turn	Q =; L = mean $L_n/ r_n $; E = 0; W = 0							
7 Row 6 without the last third of each turn	vs. row 6: Q $\downarrow$; L =; E $\downarrow$ (below 0); W = 0							
8 Turns merged across a speaker change (VAD)	Q $\downarrow$; E $\uparrow$; W $\uparrow$; more empty turns							
9 Output of 10% of turns dropped	Q $\downarrow$; L, E $\approx$ row 1; 10% empty turns							
10 Rows 1, 3–7, 9 with gold turn IDs: largest change	0 for every metric							

to 2.576 † and 1.421 † s, presumably because resegmentation spreads the untranslated words over neighbouring turns; and without turn IDs, the current scorer counts none of these words as wrong. These two failures motivate the matched-word latency, the rule to show latency only next to quality, length ratio and empty turns, and the wrong-direction detector. In addition, five unit tests check normalisation, resegmentation, LAAL, oracle versus resegmented scoring and the command-line interface on toy sessions; for instance, an oracle hypothesis scores BLEU 100.0 with a mean EndOffset of 0.0 s, and a two-word turn reaches chrF 100.0. [Draft: re-run rows 1–2 under the final protocol and record how the row-2 hypothesis is time-stamped]

C.6. Rank agreement and sim-to-real validity

Several choices could change our conclusions without changing any system: synthetic instead of real speech, the TTS engine, silver instead of human references, the random render, and the metric. Table 11 measures each by Kendall’s τ_b between two rankings of the same systems at their main operating points. Pooling wrappers and directions would inflate agreement, as their large differences recur in both rankings, so we compute τ_b within each combination of direction and wrapper and average it over these strata; we report the number n of ranked configurations and a 95% CI from 2,000 bootstrap resamples of sessions, drawn independently in each set when two sets are compared (Appendix H).

Synthetic versus real speech. The main check ranks all systems that run on both sets by chrF on CST-Bench-TAXI and on CST-Bench-Syn En↔De, both under L0 natural, so that only the speech and the dialogues differ (Figure 7); as our claims rest on oracle–realistic gaps, it also ranks the systems by their gap on each set, within each direction. We fix a threshold τ^* ([TBD]) before the test runs: if the lower CI bound reaches τ^* , CST-Bench-Syn may rank systems on its own; otherwise we report the disagreement as a finding, use CST-Bench-Syn only for contrasts within one set, such as oracle versus realistic wrappers or L0 versus L1, and test whether a telephone-band rendering of CST-Bench-Syn restores agreement. The commercial and the open TTS edition are compared in the same way on a subset of CST-Bench-Syn.

References and renders. To test how far silver references can rank systems, we rescore CST-Bench-Syn En↔De against silver references produced like the Korean ones (Appendix B) instead of the human references. To test render noise, we re-render CST-Bench-TAXI L0 natural with seeds 1–3 under new configuration names and compare the standard deviation of chrF and EndOffset across seeds with the bootstrap CI width; Table 11 lists the smallest pairwise τ_b among the three renders.

Table 11. Validity of system rankings (TAXI: CST-Bench-TAXI; Syn: CST-Bench-Syn; all L0 natural). Each row ranks the same systems at their main operating points in two ways; Kendall’s τ_b is computed within each direction $\times$ wrapper stratum (each direction for gaps) and averaged over strata, with a 95% CI from 2,000 session-bootstrap resamples; n counts the ranked configurations. For render seeds, τ_b is the smallest pairwise value among the three renders; Appendix C also compares their standard deviation with the CI width. Blank cells are pending.

Comparison	Configurations compared	n	Kendall’s τ_b	95% CI
<i>Speech, construction and rendering (chrF rankings)</i>				
TAXI vs. Syn En $\leftrightarrow$ De	those run on both sets			
oracle–realistic gap	same, ranked by gap			
Commercial vs. open TTS edition	Syn subset			
Gold vs. silver references	Syn En $\leftrightarrow$ De			
Render seeds 1–3 (chrF and EndOffset)	TAXI			
<i>Metric choice</i>				
StreamLAAL vs. EndOffset	TAXI			
StreamLAAL vs. LongYAAL	TAXI			
COMET vs. chrF	TAXI			

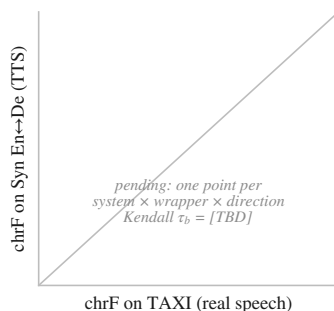


Figure 7. Sim-to-real agreement: chrF of each system, wrapper and direction at its main operating point on CST-Bench-TAXI (real speech) and on CST-Bench-Syn En $\leftrightarrow$ De (TTS), both under L0 natural. Hollow markers: realistic wrapper; grey line: identity. Kendall’s τ_b , averaged over direction and wrapper strata, is annotated in the panel; its bootstrap CI is in Table 11. [Draft: placeholder; the points appear once the baselines have run on both sets]

Metrics. StreamLAAL and EndOffset measure different waits, so a low τ_b between them is a finding rather than a validity failure: it shows that the listener’s wait reorders systems. We keep StreamLAAL as the main latency metric because it was the latency metric of the IWSLT 2025 simultaneous track (Abdulmumin et al., 2025) and shares the resegmentation of our quality metrics. LongYAAL (Polák et al., 2025), computed on SoftSegmenter resegmentation in OmniSTEval and designed against the segmentation-related bias of resegmented latency, was the primary latency metric of IWSLT 2026, which reported StreamLAAL for comparison (Adelani et al., 2026); we report how well the two agree. Rankings by string metrics alone can mislead (Kocmi et al., 2021), which favours COMET, but the TAXI references are lowercase and unpunctuated. COMET is therefore primary on CST-Bench-TAXI only if the lower CI bound of τ_b between its ranking and that of chrF reaches τ^* ; otherwise chrF is. [Draft: planned: every row of Table 11 and the points of Figure 7]

D. Baselines and Wrappers

Table 12 lists the translation systems of Section 6. All of them are scored by `cstbench eval` from committed, time-stamped pieces (Appendix C); this appendix describes how each system turns a session into pieces, how its latency settings are chosen, and which prompts the LLM-based systems use. We use public checkpoints without fine-tuning on CST data; the only tuning data is the CST-Bench-Syn calibration split, and nothing is tuned on CST-Bench-TAXI. Seed LiveInterpret 2.0 (Cheng et al., 2025), a closed system, covers only Chinese–English, which CST-Bench lacks, so we discuss it but do not run it. [Draft: implemented: the offline systems with oracle input, whose CST-Bench-TAXI scores are preliminary; planned: the streaming variant with oracle input, both wrappers, the streaming baselines, the decoupled twin and the operating points]

Offline systems with oracle input. Each gold turn is cut from `mix.wav` at its reference interval, 0.1 s margins included, and translated after it ends, one turn per prompt and without dialogue history. Qwen3-ASR-1.7B (Shi et al., 2026) tran-

Table 12. Baseline configurations, named as in Tables 3 and 6 (LLM: Qwen3.8-27B). Params: recognition plus translation models (B: billions; MODEL: total; its encoder comes from Nemotron 3.5 ASR Streaming 0.6B, which has 600M parameters per its model card and 638M as measured, RNN-T decoder included). Directions: En $\leftrightarrow$ De on CST-Bench-TAXI and, for systems with a Korean entry, on CST-Bench-Syn; Ko $\leftrightarrow$ En on CST-Bench-Syn only. Policy: when text is committed; LA-2: LocalAgreement-2. Latency knob: the setting swept on the CST-Bench-Syn calibration split; its two values (low and high latency) are chosen by the rule of Appendix D and are pending. Wrappers: O oracle segmentation, R realistic (VAD and language-ID routing), – none (the system handles the whole mix). Not listed: the diarization cascades of Table 5, which do not translate (Appendix D). *Optional; InfiniSST only if its weights are public; Hibiki-Zero is scored on its text output. Blank cells are pending.

System	Checkpoint / version	Params	Directions	Policy	Latency knob (two points)	Wrappers	Licence
<i>Offline, oracle input: gold turns, each translated after it ends</i>							
Gold $\rightarrow$ LLM	Qwen3.8-27B	27B	En $\leftrightarrow$ De Ko $\leftrightarrow$ En	turn end	–	O	Apache-2.0
ASR $\rightarrow$ LLM	Qwen/Qwen3-ASR-1.7B; Qwen3.8-27B	1.7B + 27B	En $\leftrightarrow$ De Ko $\leftrightarrow$ En	turn end	–	O	Apache-2.0
Whisper ST	openai/whisper-large-v3		De $\rightarrow$ En Ko $\rightarrow$ En	turn end	–	O	
<i>Streaming, oracle input: gold transcripts with forced-aligned word times; non-causal unit choice</i>							
Gold $\rightarrow$ LLM (MU2)	Qwen3.8-27B	27B	En $\leftrightarrow$ De	MU2 units	one point (chrF $\geq$ 50)	O	Apache-2.0
<i>Pipelines built from single-direction systems</i>							
Consecutive cascade	Silero VAD; LID; Qwen3-ASR-1.7B; Qwen3.8-27B	1.7B + 27B	En $\leftrightarrow$ De Ko $\leftrightarrow$ En	VAD endpoint	endpoint silence	R	
SeamlessStreaming			En $\leftrightarrow$ De Ko $\leftrightarrow$ En	EMMA	decision threshold	O, R	
2 $\times$ SeamlessStreaming	as above		En $\leftrightarrow$ De	EMMA	as above	–	
Streaming cascade	nvidia/nemotron-3.5- asr-streaming-0.6b; Qwen3.8-27B	0.6B + 27B	En $\leftrightarrow$ De Ko $\leftrightarrow$ En	LA-2	ASR chunk, update step	O, R	OpenMDW-1.1; Apache-2.0
M4T v2 + AlignAtt	SeamlessM4T v2		En $\leftrightarrow$ De	AlignAtt	frames f	O, R	
Canary-1B-v2	Canary-1B-v2	1B	En $\leftrightarrow$ De	AlignAtt	frames f	O, R	
+ AlignAtt*			De $\rightarrow$ En	learned	chunk size	O, R	
StreamSpeech			De $\rightarrow$ En	learned		O, R	CC BY-NC-SA 4.0
Hibiki-Zero*	hibiki-zero-3b- pytorch-bf16	3B	De $\rightarrow$ En	learned		O, R	
InfiniSST*			En $\rightarrow$ De	learned (LLM)		O, R	
<i>Controls in the ablations (Table 6)</i>							
Decoupled twin	MODEL M1; separate 1.7B-thinker pass		En $\leftrightarrow$ De	MU2 units, delay δ_t	δ_t	–	OpenMDW-1.1; Apache-2.0
Same-backbone cascade (M1 $\rightarrow$ LLM)	MODEL M1; Qwen3.8-27B	2,336M + 27B	En $\leftrightarrow$ De	LA-2	update step	–	OpenMDW-1.1; Apache-2.0
Size-matched cascade (ASR $\rightarrow$ 1.7B LLM)	Qwen3-ASR-1.7B; LLM of $\approx$ 1.7B	1.7B + $\approx$ 1.7B	En $\leftrightarrow$ De	LA-2	update step	R	
<i>Ours</i>							
MODEL	Nemotron 3.5 encoder; Qwen3-ASR-1.7B thinker	2,336M	En $\leftrightarrow$ De Ko $\leftrightarrow$ En	MU2 units, delay δ_t	δ_t	–	OpenMDW-1.1; Apache-2.0

scribes the turn with its language given, and Qwen3.8-27B translates this transcript, or the gold transcript, from the user message

```
Translate this {src_lang} utterance from a phone conversation into {tgt_lang}.
Output only the {tgt_lang} translation.
```

followed by a blank line and the turn’s text; the placeholders are English names of the languages, such as German. Decoding is greedy with at most 160 new tokens and the model’s reasoning mode switched off, and the first line of the answer is kept. Whisper-large-v3 (Radford et al., 2023) translates German turns, and on CST-Bench-Syn Korean turns, into English with its translation task and the source language given; it cannot translate into German. All models process batches of 16 turns. Every piece carries its turn ID and is stamped at the turn end e_n , so on the ideal clock EndOffset is 0 and StreamLAAL equals L_n (Appendix C). The preliminary CST-Bench-TAXI scores of these systems were computed in the scorer’s oracle mode, by turn ID and without resegmentation, and will be re-scored with resegmentation under the final protocol. We do not call these systems bounds: they translate without dialogue history, and on De→En Whisper ST already scores above Gold→LLM in BLEU (Table 3). The computation-aware clock adds the per-turn compute, measured as the time of a batch divided by its size and therefore approximate; Table 15 re-measures it without batching. The WER of the ASR transcripts is computed after the scorer’s normalisation.

Streaming variant with oracle input. To show what the commit policy costs and saves when recognition and segmentation are perfect, Gold→LLM (MU2) reveals gold transcripts with forced-aligned word times (Qwen3-ForcedAligner; Shi et al., 2026) word by word to Qwen3.8-27B, which commits translations at verified meaning units (MU2; Section 5), accepting a unit if its chrF against the aligned part of the full translation is at least 50. A unit is stamped when its last source word ends, and the rest of a turn is flushed at the turn end. As units are chosen with the whole turn in view, this system is non-causal; it runs only under the oracle wrapper and so never becomes the reference pipeline P of Section 6.1. *[Draft: planned; the German minimum unit is set on held-out data before the run]*

Oracle wrapper. A single-direction system runs as two instances, one per target language. The wrapper cuts each gold turn $[b_n, e_n]$ from the mono mix, so that under L1 the cut also contains the overlapping speech of the other speaker, and streams it in real time to the instance that translates into the other speaker’s language. Instances are reset at every turn and receive no dialogue history. A piece emitted after the instance has consumed u seconds of the turn is stamped $b_n + u$ on the ideal clock, plus compute time on the computation-aware clock, and carries the turn ID. The main scores are computed with the turn IDs removed, so that the scorer resegments oracle-wrapper output like any other; scoring with them (oracle mode, Appendix C) is a diagnostic of how much resegmentation alone moves the scores. Cutting the turn from its speaker’s channel of 2ch.wav instead, in the channel order recorded in the timeline, gives the oracle-separation variant of Table 4. *[Draft: planned]*

Realistic wrapper. The realistic wrapper receives only the mix and the language pair. Silero VAD (Silero Team, 2024) segments the stream online: a segment opens at a detected speech onset and closes after a minimum silence. A spoken-language identifier restricted to the session’s two languages labels each segment from its onset and updates the label as audio arrives. The segment is streamed to the instance that translates into the other language and is re-routed to the other instance if the label flips; text already committed stays. Pieces are stamped from the segment onset as in the oracle wrapper. As identifier we use the automatic language detection of Nemotron 3.5 ASR (NVIDIA, 2026), or the language identification of Qwen3-ASR (Shi et al., 2026) if the former proves less accurate on the calibration split. The VAD’s speech-probability threshold, minimum silence and speech padding and the identifier’s settings are fixed once on that split, for all systems. As the split is clean 16 kHz TTS, these settings may suit telephone-band CST-Bench-TAXI less well (Appendix B). A short minimum silence splits turns at pauses, whereas a long one merges turns across the short L0 natural gaps (median drawn gap 0.223 s on CST-Bench-TAXI, to which the 0.1 s turn margins add up to 0.2 s of silence; Section 4); under L1, the VAD hears one continuous segment across the speaker change, so the second turn inherits the first turn’s route unless the label flips. Sweeps of the minimum silence on CST-Bench-TAXI show how sensitive the realistic wrapper is to this setting; they serve analysis only, not tuning. The factorial of Table 4 combines components of the two wrappers: gold turns with the identifier, and VAD segments with the gold language of the turn that overlaps each segment most. *[Draft: planned; the identifier is chosen on the calibration split]*

Streaming systems. SeamlessStreaming (Seamless Communication et al., 2023) translates speech into a requested language with Efficient Monotonic Multihead Attention (EMMA), a variant of monotonic multihead attention (Ma et al.,

2020b), and runs behind both wrappers. Two SeamlessStreaming instances on the whole mix, one per target language and without routing, form the naive bidirectional baseline: each instance also receives speech that is already in its target language, and what it emits for that speech counts as wrong-direction output (copy-through or wrong language; Appendix C). The streaming cascade passes the growing transcript of Nemotron 3.5 ASR (NVIDIA, 2026), a cache-aware streaming RNN-T, to Qwen3.8-27B, which re-translates it after every update, with the committed translation as a forced prefix and the instruction of the offline prompt, and commits the longest common prefix of its last two translations (LocalAgreement-2; Liu et al., 2020; Polák et al., 2022; Macháček et al., 2023). It follows recent cascades that translate a growing transcript with an LLM (Fuxa & Macháček, 2026), but its policy needs no access to attention weights. SeamlessM4T v2 (Seamless Communication et al., 2023; SEAMLESS Communication Team et al., 2025) and, optionally, Canary-1B-v2 (Sekoyan et al., 2025) are offline speech translation models that we run simultaneously in simulstream (Gaido et al., 2025) with AlignAtt (Papi et al., 2023b), which emits a token only if the audio frame it attends to most is not among the last f frames. StreamSpeech (Zhang et al., 2024), which learns translation and policy jointly, covers De→En; InfiniSST (Ouyang et al., 2025b), an LLM-based system for unbounded speech, covers En→De and is included only if its weights are public. Hibiki-Zero (Labiausse et al., 2026), a simultaneous speech-to-speech and speech-to-text translation model trained without word-level aligned data, translates German among other languages into English and has public weights (CC BY-NC-SA 4.0); it is an optional De→En baseline, scored on its text output only. *[Draft: planned; checkpoints and parameter counts left blank in Table 12 are filled when the runs are set up]*

Cascades. The consecutive cascade is the realistic counterpart of the offline cascade: it waits for the VAD endpoint of each segment, identifies the segment’s language and then runs Qwen3-ASR-1.7B and Qwen3.8-27B with the offline prompt, so its EndOffset includes the endpoint silence and the compute time. In Table 6, the same-backbone cascade feeds the transcripts of M1, MODEL’s stage-2 checkpoint (Section 5), to the LocalAgreement-2 translation stage, and the size-matched cascade pairs Qwen3-ASR-1.7B with an LLM of about 1.7B parameters, the size of MODEL’s language model. The diarization cascades of Table 5 combine pyannote (Bredin, 2023) with offline Qwen3-ASR-1.7B, and Streaming Sortformer (Medennikov et al., 2025) with streaming Nemotron 3.5 ASR. *[Draft: planned]*

Decoupled twin. The decoupled twin of Table 6 tests the coupling of turn-taking and translation at MODEL’s scale and commit policy. It starts from the same M1 checkpoint and 1.7B thinker as MODEL: M1 writes the speaker-tagged transcript, whose speaker tags fix turn boundaries and directions as hard decisions, and a separate translation pass, trained on the same data, then translates each fixed segment with the same MU2 commit policy and δ_t . Translation therefore cannot revise segmentation or routing, while scale, data, simulated-session exposure and commit policy match MODEL; the same-backbone cascade shares the M1 exposure but not the translator’s scale or policy. *[Draft: planned; the twin needs its own Stage-3 run]*

Operating points. Each system with a latency knob sweeps it over three to four settings, which gives the curves of Figure 4, and runs in the tables at two operating points, chosen once on the CST-Bench-Syn calibration split (about 5% of its sessions) by a rule that is fixed before the test runs and is the same for all systems. Budgets are relative to turn length, because CST-Bench-TAXI turns are short (median 2.86 s) and differ in length between directions, so an absolute budget would mean different things per direction: with $\bar{D}$ the mean source-turn duration of a direction on the calibration split, the *simultaneous* budget is $\text{StreamLAAL} \leq 0.5 \bar{D}$ (the main operating point) and the *consecutive* budget is $\text{StreamLAAL} \leq \bar{D}$, the lag of a system that translates each turn after it ends. We sweep the knob under the oracle wrapper, so that one setting serves both wrappers, and for each budget take the setting with the highest COMET (chrF for Korean targets) that stays within it; a system that cannot meet a budget runs at its fastest setting, which we mark. The consecutive cascade, which exists only in realistic form, sweeps the minimum silence of its own VAD instead, and MODEL’s two operating points are values of δ_t chosen by the same rule. The offline systems and their streaming variant with oracle input have a single operating point. *[Draft: planned; until CST-Bench-Syn is built, systems run at their documented default settings and are re-run once the calibration split exists]*

Release. We release the wrapper code with the scorer, including segmentation, language identification, routing and the settings of every operating point, so that any single-direction system can be evaluated under the same protocol. For CST-Bench-TAXI, we release per-turn scores keyed by turn ID but no system outputs (Appendix H).

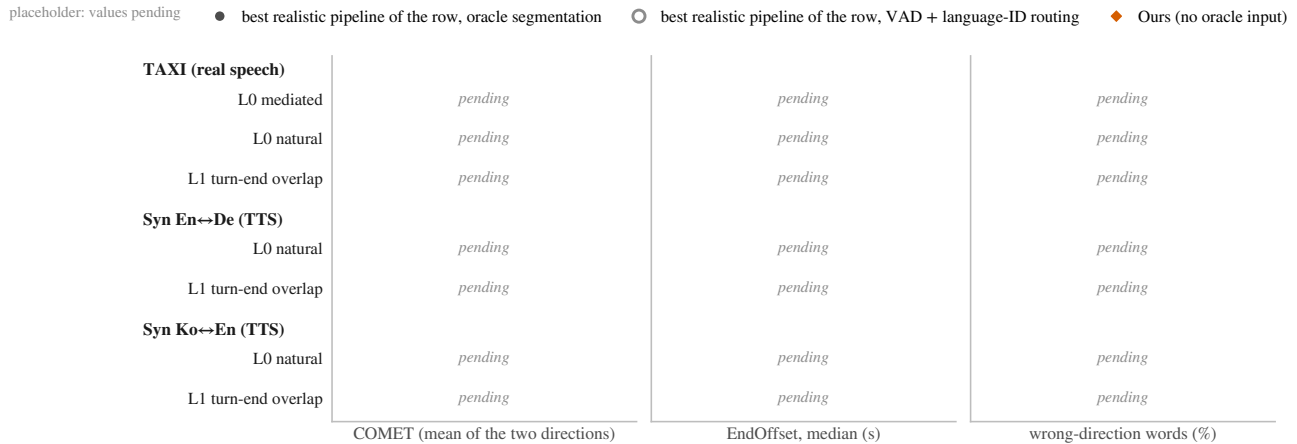


Figure 8. Where the conversational difficulty lies (values pending; numbers will be in Table 13). For each timing condition and language pair: the main pipeline with the highest realistic-wrapper COMET in that row (named at the left), with oracle turn segmentation (filled) and with VAD and language-ID routing (hollow), and MODEL (diamond; no oracle input), each at its main operating point. Panels: COMET (mean of the two directions), median EndOffset and wrong-direction words. Rows are compared only through their gaps, not through absolute COMET.

E. Additional Results

This appendix gives the oracle gap by timing condition and language pair (Figure 8) with the full numbers behind it and behind Table 3 (Table 13), the decomposition of the gap, a dose-response analysis over the inter-turn gap (Figure 9), latency by source-turn duration (Figure 10), computation-aware latency (Table 15), a failure taxonomy (Table 16), further transcript and speaker metrics, and the variants of Table 6. All analyses follow the protocol of Section 4: 95% session-clustered bootstrap intervals, and no threshold or operating point tuned on TAXI. So far, only the offline rows on CST-Bench-TAXI L0 natural (preliminary) and a few timings exist. [Draft: every other cell is pending; L1, the fixed-gap renders and CST-Bench-Syn are planned]

Gap by timing condition and language pair. Figure 8, summarised in Section 6.3, contrasts for every timing condition and language pair the main pipeline with the highest realistic-wrapper COMET in that row, under the oracle and the realistic wrapper, with MODEL, which receives no oracle input. Both wrappers of a row use the same pipeline, so each gap compares the same system with and without oracle segmentation and routing; the gaps of all main pipelines are in Table 14. COMET is averaged over the two directions; as one side of every synthetic pair is translationese (Appendix B), rows are compared only through their gaps, never through absolute COMET.

Full grid. For every timing condition, language pair and direction, Table 13 lists the systems run under it, each at its main operating point, fixed before the test runs; part (a) also repeats every row of Table 3, so that each has its interval. The offline systems receive gold turns and translate each turn after it ends. Their StreamLAAL therefore equals the mean source-turn duration (4.624 s for En→De and 3.086 s for De→En on CST-Bench-TAXI), and their EndOffset is 0 on the ideal clock because their output is time-stamped at the gold turn end; Table 15 adds their compute. The gold-transcript row isolates translation quality, the Qwen3-ASR row adds the recognition of telephone-band speech, and Whisper ST covers De→En only. Under L1, an oracle segment cut from the mono mix also contains the overlapping speech of the other speaker; the oracle-separation row of Table 4 removes it. On CST-Bench-Syn we interpret only rankings and effects relative to each pair’s own offline systems with oracle input, never absolute scores across pairs.

Decomposing the gap. We decompose the gap of every main pipeline on every set rather than of one selected pipeline (Table 14). For one pipeline and set, let Q_{TL} be the COMET of its row with gold turns ($T=1$) or VAD segments ($T=0$) and with the gold ($L=1$) or the identified ($L=0$) source language, so that its oracle gap (Section 6.1) is $G = Q_{11} - Q_{00}$. The segmentation effect

$$E_T = \frac{1}{2} [(Q_{11} - Q_{01}) + (Q_{10} - Q_{00})]$$

Table 13. Full results by timing condition and language pair, the numbers behind Figures 8 and 9 (ideal clock; each system at its main operating point, fixed before the test runs). Part (a) covers CST-Bench-TAXI and repeats every row of Table 3; parts (b) and (c) follow. Systems and Seg. as in Table 3; X is German on CST-Bench-TAXI and CST-Bench-Syn En $\leftrightarrow$ De, and Korean on CST-Bench-Syn Ko $\leftrightarrow$ En. Latencies in seconds over non-empty turns: mean StreamLAAL and median EndOffset per direction; wrong-direction words, median switch latency and empty turns are pooled over both directions and do not apply to single-direction systems; Table 15 lists each system’s encoder look-ahead ℓ_{enc} . Each estimate will carry the half-width of its 95% session-clustered bootstrap CI as a subscript. Offline systems translate each oracle turn after it ends, so their StreamLAAL is the mean source-turn duration and their EndOffset is 0 by construction. [†]Preliminary, scored by gold turn ID without resegmentation (Appendix C); to be re-run under the final protocol (same runs as Table 3; no CIs yet). –: not applicable; blank: pending; [°]optional. [Draft: L1 is not rendered yet]

System	Seg.	En→X				X→En				Both directions		
		COMET	chrF	Stream-LAAL	End-Offset	COMET	chrF	Stream-LAAL	End-Offset	Wrong dir. (%)	Switch latency	Empty turns (%)
<i>(a) CST-Bench-TAXI, LO natural</i>												
Gold→LLM	oracle		71.72 [†]	4.62 [†]	0		58.05 [†]	3.09 [†]	0			
ASR→LLM	oracle		67.97 [†]	4.62 [†]	0		56.61 [†]	3.09 [†]	0			
Whisper ST	oracle	—	—	—	—		56.67 [†]	3.09 [†]	0	—	—	—
Gold→LLM (MU2)	oracle											
Consecutive cascade	realistic											
SeamlessStreaming	oracle											
SeamlessStreaming	realistic											
2×SeamlessStreaming	none											
Streaming cascade	oracle											
Streaming cascade	realistic											
M4T v2 + AlignAtt	oracle											
M4T v2 + AlignAtt	realistic											
StreamSpeech	oracle	—	—	—	—					—	—	—
StreamSpeech	realistic	—	—	—	—					—	—	—
Hibiki-Zero [°]	oracle	—	—	—	—					—	—	—
Hibiki-Zero [°]	realistic	—	—	—	—					—	—	—
InfiniSST [°]	oracle					—	—	—	—	—	—	—
InfiniSST [°]	realistic					—	—	—	—	—	—	—
MODEL M2	none											
MODEL M3, low δ_t	none											
MODEL M3, high δ_t	none											
<i>CST-Bench-TAXI, LO mediated</i>												
Gold→LLM	oracle											
ASR→LLM	oracle											
Whisper ST	oracle	—	—	—	—					—	—	—
Consecutive cascade	realistic											
SeamlessStreaming	oracle											
SeamlessStreaming	realistic											
Streaming cascade	oracle											
Streaming cascade	realistic											
M4T v2 + AlignAtt	oracle											
M4T v2 + AlignAtt	realistic											
MODEL (M3, main δ_t)	none											

Table 13. (continued) Part (a), CST-Bench-TAXI under L1, and part (b), CST-Bench-Syn En↔De, with the columns and conventions above. [Draft: L1 and CST-Bench-Syn are planned]

System	Seg.	En→X				X→En				Both directions		
		COMET	chrF	Stream-LAAL	End-Offset	COMET	chrF	Stream-LAAL	End-Offset	Wrong dir. (%)	Switch latency	Empty turns (%)
<i>(a, continued) CST-Bench-TAXI, L1 (turn-end overlap)</i>												
Gold→LLM	oracle											
ASR→LLM	oracle											
Whisper ST	oracle	–	–	–	–					–	–	–
Consecutive cascade	realistic											
SeamlessStreaming	oracle											
SeamlessStreaming	realistic											
Streaming cascade	oracle											
Streaming cascade	realistic											
M4T v2 + AlignAtt	oracle											
M4T v2 + AlignAtt	realistic											
MODEL (M3, main δ_t)	none											
<i>(b) CST-Bench-Syn En↔De, L0 natural</i>												
Gold→LLM	oracle											
ASR→LLM	oracle											
Whisper ST	oracle	–	–	–	–					–	–	–
Consecutive cascade	realistic											
SeamlessStreaming	oracle											
SeamlessStreaming	realistic											
Streaming cascade	oracle											
Streaming cascade	realistic											
MODEL (M3, main δ_t)	none											
<i>CST-Bench-Syn En↔De, L1 (turn-end overlap)</i>												
Gold→LLM	oracle											
ASR→LLM	oracle											
Whisper ST	oracle	–	–	–	–					–	–	–
Consecutive cascade	realistic											
SeamlessStreaming	oracle											
SeamlessStreaming	realistic											
Streaming cascade	oracle											
Streaming cascade	realistic											
MODEL (M3, main δ_t)	none											

Table 13. (continued) Part (b), CST-Bench-Syn Ko↔En, and part (c): CST-Bench-TAXI re-rendered with one fixed gap at every speaker change (a negative gap starts the next turn before the previous one ends), the numbers behind Figure 9, for the reference pipeline P (Section 6.1) under both wrappers and MODEL; COMET is the mean of the two directions and EndOffset the median over both. Blank: pending. [Draft: CST-Bench-Syn and the fixed-gap renders are planned]

System	Seg.	En→X				X→En				Both directions		
		COMET	chrF	Stream-LAAL	End-Offset	COMET	chrF	Stream-LAAL	End-Offset	Wrong dir. (%)	Switch latency	Empty turns (%)
<i>(b, continued) CST-Bench-Syn Ko↔En, L0 natural</i>												
Gold→LLM	oracle											
ASR→LLM	oracle											
Whisper ST	oracle	–	–	–	–					–	–	–
Consecutive cascade	realistic											
SeamlessStreaming	oracle											
SeamlessStreaming	realistic											
Streaming cascade	oracle											
Streaming cascade	realistic											
MODEL (M3, main δ_t)	none											
<i>CST-Bench-Syn Ko↔En, L1 (turn-end overlap)</i>												
Gold→LLM	oracle											
ASR→LLM	oracle											
Whisper ST	oracle	–	–	–	–					–	–	–
Consecutive cascade	realistic											
SeamlessStreaming	oracle											
SeamlessStreaming	realistic											
Streaming cascade	oracle											
Streaming cascade	realistic											
MODEL (M3, main δ_t)	none											

Table 14. Full oracle-component factorial for every main pipeline at its main operating point (ideal clock). ✓: oracle T (gold turns, else VAD segments), L (gold language, else spoken-language identification restricted to the pair) or S (each speaker’s own channel; L1 only). Row 4 is the realistic wrapper. TAXI: CST-Bench-TAXI; Syn: CST-Bench-Syn. COMET (mean of both directions) and wrong-direction words (WD, %); the effects E_T , E_L , I and E_S and their CIs follow from these rows (Appendix E). Blank: pending; –: not applicable.

		Oracle			TAXI L0 nat.		TAXI L1			Syn En↔De L0		Syn En↔De L1		Syn Ko↔En L1	
Pipeline	#	T	L	S	COMET	WD	COMET	WD	Switch lat. (s)	COMET	WD	COMET	WD	COMET	WD
Consecutive cascade	1	✓	✓	✗											
	2	✓	✗	✗											
	3	✗	✓	✗											
	4	✗	✗	✗											
	5	✓	✓	✓		–		–			–		–		
SeamlessStreaming	1	✓	✓	✗											
	2	✓	✗	✗											
	3	✗	✓	✗											
	4	✗	✗	✗											
	5	✓	✓	✓		–		–			–		–		
Streaming cascade	1	✓	✓	✗											
	2	✓	✗	✗											
	3	✗	✓	✗											
	4	✗	✗	✗											
	5	✓	✓	✓		–		–			–		–		
MODEL (no oracle input)															

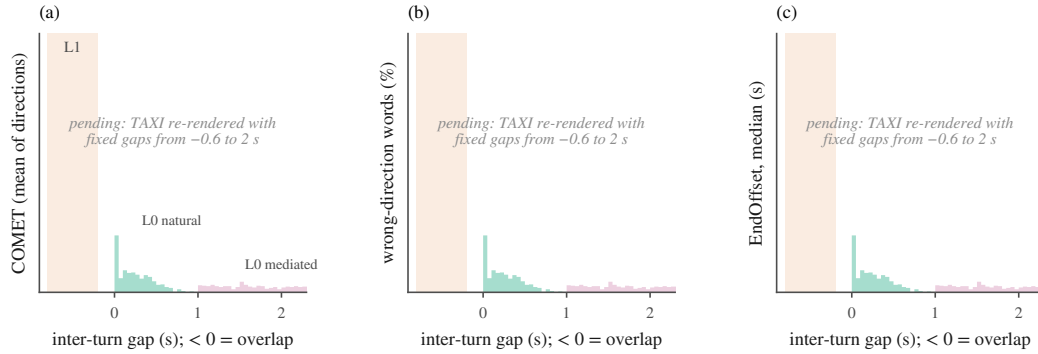


Figure 9. Gap dose-response: CST-Bench-TAXI re-rendered with one fixed gap at every speaker change, from -0.6 s (overlap) to 2 s. (a) COMET (mean of the two directions), (b) wrong-direction words and (c) median EndOffset for the reference pipeline P (solid: oracle segmentation; dashed: VAD and language-ID routing) and MODEL. Background histograms show the gaps drawn at the 540 speaker changes under L0 natural (green) and L0 mediated (pink); the shaded band is the overlap range of L1, whose next turn starts 0.2 – 0.8 s before the previous one ends. Numbers in part (c) of Table 13. [Draft: results pending; the fixed-gap renders are planned]

and the routing effect $E_L = \frac{1}{2}[(Q_{11} - Q_{10}) + (Q_{01} - Q_{00})]$ each average the loss from replacing one oracle component over both settings of the other component. They sum to G , which makes them the Shapley attribution for two factors, and the interaction $I = Q_{11} - Q_{10} - Q_{01} + Q_{00}$ measures how far the two losses compound. Under L1, the separation effect $E_S = Q_{11S} - Q_{11}$ is the COMET gained when each oracle segment is cut from its speaker’s own channel instead of the mix (row 5 vs. row 1). We report E_T , E_L , I and E_S in COMET points with session-bootstrap CIs, not as shares of G , which are unstable when G is small; Table 4 summarises them and Table 14 lists every row.

Gap dose-response. Each timing condition draws its gaps from one distribution. To test whether pipelines degrade smoothly as turns follow each other more closely, we re-render CST-Bench-TAXI with one fixed gap g at every speaker change, $g \in \{-0.6, -0.4, -0.2, 0.05, 0.2, 0.5, 1.0, 2.0\}$ s, each rendering under its own configuration name and check-sums; a negative gap starts the next turn $|g|$ before the previous one ends, as in L1. We run every main pipeline with both wrappers, and MODEL (Figure 9; numbers in part (c) of Table 13). A smooth trend would show that the conclusions do not hinge on the chosen L0 and L1 distributions; a sharp drop would locate the gap at which VAD-based routing breaks down. From $g = 2$ s to $g = -0.6$ s, COMET changes by [TBD] with the realistic wrapper, by [TBD] with oracle segmentation and by [TBD] for MODEL.

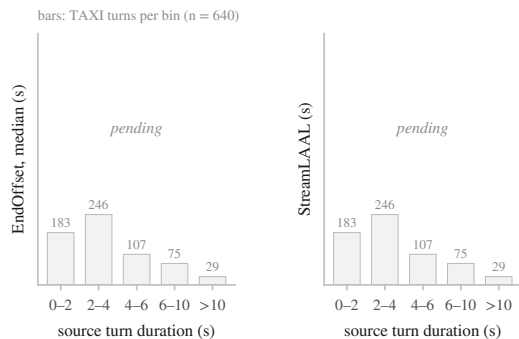


Figure 10. Why simultaneous? Median EndOffset (left) and StreamLAAL (right) by source-turn duration on CST-Bench-TAXI L0 natural, for the realistic consecutive cascade, the best realistic streaming pipeline and MODEL. Bars: CST-Bench-TAXI turns per duration bin (183, 246, 107, 75 and 29 of 640). [Draft: results pending]

Why simultaneous? Latency by turn length. With oracle endpoints, offline systems reach an EndOffset of 0 on the ideal clock, which raises the question of why simultaneity is needed at all. A deployed consecutive system must first detect that a turn has ended, so the listener waits for the endpoint detector and, on the computation-aware clock, also for the translation of the whole turn. Lag within a turn grows with the turn as well: a system that emits a turn’s translation only after the turn ends scores a StreamLAAL of at least the turn’s duration (Section 4). CST-Bench-TAXI turns are short (median 2.86 s), but 22.5% of them exceed 5 s (Table 7). Figure 10 therefore breaks median EndOffset and StreamLAAL down by source-turn duration for the realistic consecutive cascade, the best realistic streaming pipeline and MODEL. For turns longer than 6 s, the median EndOffset is [TBD] s for the consecutive cascade, [TBD] s for the best streaming pipeline and [TBD] s for MODEL.

Computation-aware latency. Main-text latencies use the ideal clock. Table 15 times every system on one H200 GPU, as a single stream without batching, and re-scores it on the computation-aware clock, on which an emission time also counts the processing time spent until the emission (Ma et al., 2020c). It includes the three controls of Table 6, because the ideal clock hides how much their compute differs from MODEL’s: the same-backbone cascade, for instance, re-translates the growing transcript with a 27B LLM after every update. Streaming systems receive the audio in 80 ms chunks; we report the median and the 99th percentile of the processing time per chunk (tick) and the real-time factor (RTF). Consecutive and offline systems are timed per turn. The only timings so far were taken outside this protocol. The single-speaker backbone with the 0.6B thinker takes about 60 ms per chunk at the median on an H200 but 100–150 ms at the 99th percentile, so it misses the 80 ms budget there; on an M4 Mac with the thinker in MLX (fp16), it takes 56 ms per chunk at the median and runs at an RTF of 0.82. In the preliminary offline run, batched compute moved the median EndOffset of the offline systems from 0 to at most about 0.2[†] s (†preliminary; to be re-run under the final protocol). The 1.7B thinker has not been timed, and we claim real time only for a system with a measured RTF below 1.

Failure taxonomy. Table 16 counts six failure types per 100 source turns; short turns are a particular risk, as 59 CST-Bench-TAXI turns have at most three words. The detectors work on the scorer’s alignments and timelines, so the table reports counts without showing any TAXI text, and their thresholds are fixed before the test runs. Merged turns, in which one VAD segment or output piece spans turns of both speakers, send one speaker’s words to the wrong channel or delay the switch. A truncated tail leaves the last words of a turn untranslated, for example when a segment is closed early. Hallucinations are output words without an aligned reference word and with low reference-free quality (Rei et al., 2022b), cross-checked against the critical-error spans of xCOMET (Guerrero et al., 2024). We estimate the precision of each detector by hand on 50 CST-Bench-Syn turns per condition, and any example we show comes from CST-Bench-Syn. Under L1, P with the realistic wrapper merges [TBD] turns per 100 (L0 natural: [TBD]), and MODEL merges [TBD].

Transcripts and speakers. Table 5 scores DER with a 0.25 s collar. Without a collar, every boundary error counts at the 80 ms resolution; DER on CST-Bench-TAXI L0 natural is then [TBD] for MODEL with $\delta_s = 2$ and [TBD] for the streaming diarization cascade, and [TBD] and [TBD] under L1. The time-constrained tcpWER of Appendix C on CST-Bench-TAXI L1 is [TBD] for MODEL with $\delta_s = 2$ and [TBD] for the streaming diarization cascade. In this release each speaker keeps one language, so the language reveals the speaker and DER mostly measures boundary timing; speakers

Table 15. Computation-aware latency and real-time factor (RTF) on one H200 GPU, single stream, no batching; CST-Bench-TAXI L0 natural, pooled over both directions. ℓ_{enc} : encoder look-ahead, counted at its bound. Tick: processing time per 80 ms input chunk; RTF: processing time over audio duration. A real-time claim needs a measured RTF below 1; a tick above 80 ms means that the output falls behind the input for a while. Ideal clock: an emission time is the audio consumed; computation-aware (CA) clock: it also counts the processing time spent until the emission (Ma et al., 2020c). StreamLAAL is a mean and EndOffset a median over non-empty turns. The batched offline row covers the three offline systems of the preliminary offline run, with approximate compute; the context rows are earlier measurements of the single-speaker backbone with the 0.6B thinker (not a CST system), taken outside this protocol. [†]preliminary; to be re-run under the final protocol. $\dashv$: not applicable or not reported; blank: pending. [Draft: planned except the batched offline row and the context rows]

System	Hardware	ℓ_{enc} (ms)	Tick (ms)		RTF	StreamLAAL (s)		EndOffset (s)	
			p50	p99		ideal	CA	ideal	CA
MODEL (1.7B thinker)	H200	80							
SeamlessStreaming	H200								
Streaming cascade	H200								
M4T v2 + AlignAtt	H200								
Consecutive cascade (unbatched)	H200	–	–	–				0	
Offline: ASR→LLM, per turn (unbatched)	H200	–	–	–				0	
Offline systems, batched	–	–	–	–	–	–	–	0	$\leq 0.2^{\dagger}$
<i>Controls of Table 6</i>									
Decoupled twin	H200	80							
Same-backbone cascade (M1→LLM)	H200	80							
Size-matched cascade (ASR→1.7B LLM)	H200								
<i>Context: single-speaker ASR backbone, 0.6B thinker (not a CST system)</i>									
0.6B backbone	H200	80	≈ 60	100–150	–	–	–	–	–
0.6B backbone, MLX fp16 thinker	M4 Mac	80	56	–	0.82	–	–	–	–

who switch languages, which we exclude, would break this equivalence.

Ablations. Every row of Table 6 starts from the same M1 checkpoint (Section 5.4). The reference, MODEL (M3), commits translations at MU2 units, writes speaker tags, sees L1-like overlapping sessions in training and is conditioned on δ_t through its prefix (Section 5.2). Each ablation changes one of these: translations committed only at turn ends (the targets of M2), direction tags without speaker tags, training from M1 without L1-like sessions (optional), and one model per δ_t instead of the conditioning (optional). Without speaker tags, the stream still carries the direction tags, which reveal the speaker as long as each speaker keeps one language; this row therefore tests whether an explicit speaker channel helps translation, not whether coupling turn-taking and translation helps, which the decoupled twin and the same-backbone cascade test (Section 6.5 and Appendix D). Rows of runs that are cut are removed, not left blank; the computation-aware latency of the three controls is in Table 15.

Table 16. Failure taxonomy: counts per 100 source turns on CST-Bench-TAXI L0 natural (L0) and L1, found by automatic detectors on the scorer’s alignments; pipelines run with the realistic wrapper. Each detector’s precision is checked by hand on 50 CST-Bench-Syn turns per condition, never on TAXI text. Consec.: consecutive cascade; Seamless: SeamlessStreaming; Stream. casc.: streaming cascade; M4T+AA: M4T v2 + AlignAtt. Thresholds are fixed before the test runs. Blank: pending. [Draft: planned]

Failure type	Detector	Consec.		Seamless		Stream. casc.		M4T+AA		MODEL	
		L0	L1	L0	L1	L0	L1	L0	L1	L0	L1
Merged turns	One VAD segment, or one output piece, spans turns of both speakers										
Wrong direction	Words on the wrong channel for the active turn, in the wrong language, or copied from the source (scorer detector)										
Truncated tail	The final words of the turn’s reference stay unaligned										
Empty turn	No output word is assigned to the turn										
Hallucination	Output with no aligned reference word and low reference-free QE (Rei et al., 2022b); xCOMET critical-error spans (Guerreiro et al., 2024) as a cross-check										
Late switch	Switch latency above 2 s										

F. Model and Backbone Details

This appendix specifies the streaming backbone of MODEL, the only part that is built, and evaluates it on single-speaker English and Korean transcription; none of its numbers is a CST result. It also records the training recipes of the planned stages. Table 17 lists the training stages, Table 18 and Figure 12 report the backbone, and Figure 11 works through the delay rule of Section 5.1 on a recorded utterance.

Encoder. The encoder of Nemotron 3.5 ASR Streaming 0.6B (NVIDIA, 2026) is a cache-aware FastConformer (Gulati et al., 2020; Rekesh et al., 2023; Noroozi et al., 2024): 128-bin log-mel features (25 ms window, 10 ms hop), depthwise-striding subsampling by 8 and 24 layers with $d_{\text{model}} = 1024$ and 8 heads, so one frame covers 80 ms and a stream of T s yields $K = \text{round}(12.5 T)$ frames. Its attention context $[56, N]$ gives the number of 80 ms frames visible to the left and to the right. We use $[56, 0]$, i.e. 4.48 s of left context and none on the right; the released checkpoint defaults to $[56, 3]$, which adds 240 ms of right context. With $[56, 0]$, encoding a whole utterance at once yields the same frames as streaming, so training encodes each batch without a cache, and live inference uses the cache-aware step, whose output is frame-identical. The encoder’s own RNN-T decoder (Graves, 2012) serves only as a baseline (Table 18). When the encoder is trained, its attention context is sampled per step from $[56, 0]$, $[56, 3]$, $[56, 6]$ and $[56, 13]$, so one checkpoint can be evaluated at several look-aheads (Figure 12).

Effective look-ahead. A nominal context can hide look-ahead, so we measure it. We encode a 14.84 s LibriSpeech (Panayotov et al., 2015) utterance from its full features and from features truncated at 3, 6, 9 and 12 s and compare the outputs frame by frame (fp32, TF32 disabled, relative tolerance 10^{-3}); the effective look-ahead is the span before the cut in which the outputs differ. With $[56, 0]$, only the last frame changes, through the boundary of the subsampling convolution, so the look-ahead is 80 ms at every cut; with $[56, 3]$, it is 160, 240 and 320 ms (minimum, mean and maximum over the cuts). Two strictly causal reference models give 0 ms, which validates the test. The test also shows why Qwen3-ASR’s own audio encoder is removed: in the implementation we tested, its attention spans the whole utterance, and even its intended mode with 1 s blocks looks ahead by 420 ms on average and by up to 800 ms.

Adapter and thinker. The adapter is LayerNorm, Linear ($1024 \rightarrow 2048$), GELU and Linear ($2048 \rightarrow h$), where h is the hidden size of the thinker: 1024 for the 0.6B thinker and 2048 for the 1.7B one (4.2M and 6.3M adapter parameters). The 0.6B thinker is a Qwen3 decoder with 28 layers, 16 query and 8 key-value heads of dimension 128, a feed-forward size of 3072, a vocabulary of 151,936 tokens and tied input and output embeddings. The adapter output replaces the input embedding at each audio placeholder position (token 151676), one per 80 ms chunk. MODEL uses the 1.7B thinker, the language model of Qwen3-ASR-1.7B (Shi et al., 2026; Xu et al., 2025); the 0.6B thinker serves for pilots. With the 1.7B thinker, Stage 1 trains 2,336M parameters: thinker, encoder and adapter. The encoder is released under OpenMDW-1.1 and the thinker under Apache-2.0.

Sequence and special tokens. The input-only prefix follows the chat format of Qwen3-ASR with an empty system turn; the assistant turn opens with the line “language English” or “language Korean”, the transcript marker and $\langle \text{DELAY}_\delta \rangle$. The prefixes of the two languages differ only in that line, so the language is an input and the backbone performs no language identification. Each chunk k then contributes a_k , the text tokens due in it and the end-of-chunk token $\langle \text{next} \rangle$ (NEXT_AUDIO in Figure 11), the analogue of a transducer blank (Graves, 2012; Seide et al., 2024). After the last chunk, a flush round ($\langle \text{empty} \rangle$), the remaining text tokens, $\langle \text{next} \rangle$ runs if tokens remain, and a final round ($\langle \text{empty} \rangle$, $\langle \text{next} \rangle$) always runs; $\langle \text{empty} \rangle$ (EMPTY_AUDIO) is an input placeholder that carries no loss. Twelve special tokens use spare rows of the embedding matrix, so the vocabulary is not resized, and each is initialised as the mean embedding plus $0.02 \mathcal{N}(0, 1)$ noise: $\langle \text{next} \rangle$ (id 151705), $\langle \text{empty} \rangle$ (151706), the speaker tags $\langle A \rangle$ and $\langle B \rangle$ (151707 and 151708), which are reserved but untrained and blocked at decoding, and $\langle \text{DELAY}_1 \rangle$ to $\langle \text{DELAY}_8 \rangle$ (151709–151716). In the 0.6B model, token ids end at 151716 while the embedding matrix has 151,936 rows, which leaves 219 unused rows for the target-language tags and the translation delay tokens of MODEL, whose prefix (Section 5.2) names the language pair and carries $\langle \text{DELAY}_\delta \rangle$ and $\langle \text{TDELAY}_\delta \rangle$. With one audio position per chunk, the longest CST-Bench-TAXI session (90.02 s) spans about 1.1k audio positions. [Draft: planned: these tags are not yet allocated]

Targets, loss and decoding. Targets are normalised lexical transcripts (lowercase, no punctuation, numbers spelled out). Word end times come from Qwen3-ForcedAligner (Shi et al., 2026) at 80 ms resolution, and all tokens of a word are due in the chunk that the delay rule assigns to the word’s end; tokens due at or after the last chunk go to the flush round,

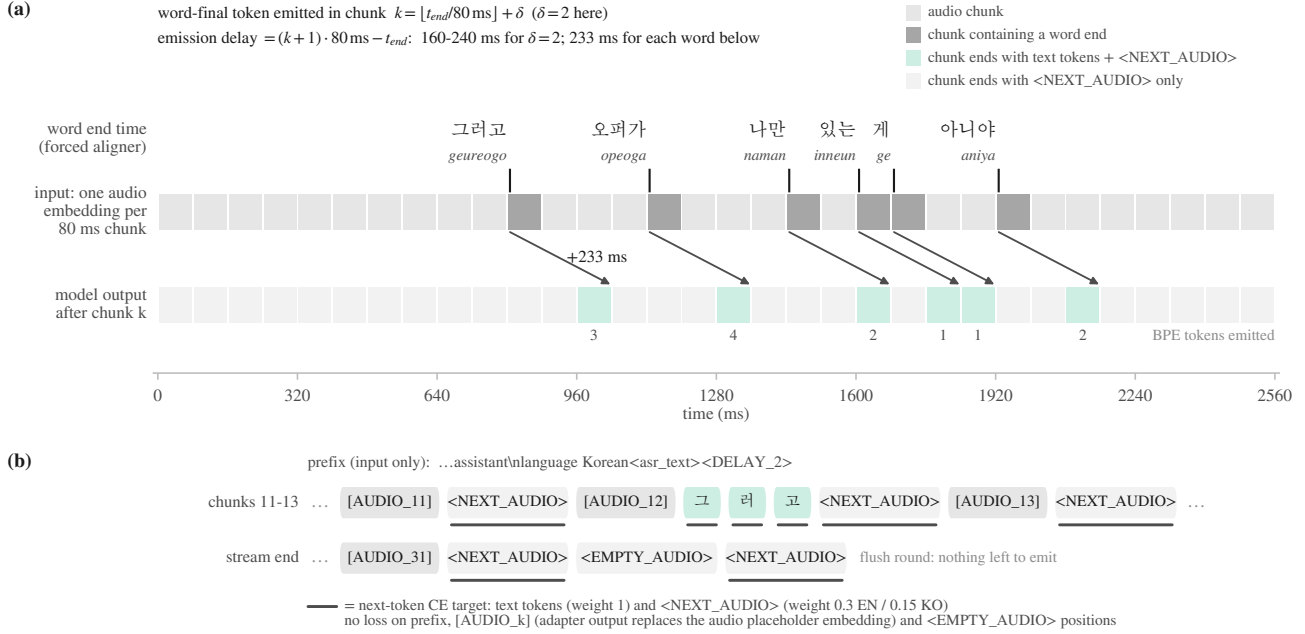


Figure 11. Delay rule of the single-speaker backbone on a recorded KsponSpeech (Bang et al., 2020) utterance (2.6 s, $K = 32$ chunks, $\delta = 2$). (a) One audio embedding enters per 80 ms chunk. A word that ends at t_{end} (forced alignment) is emitted, with all its BPE tokens, after chunk $k = \lfloor t_{\text{end}}/80 \text{ ms} \rfloor + \delta$, so its emission delay $(k + 1) \cdot 80 \text{ ms} - t_{\text{end}}$ lies between 160 and 240 ms for $\delta = 2$; here it is 233 ms for all six words. Numbers under the output row count the BPE tokens emitted. (b) Serialized LM sequence for chunks 11–13 and the end of the stream. Underlined tokens are next-token targets, weighted 1 for text and 0.3 (English) or 0.15 (Korean) for the end-of-chunk token <next> (<NEXT_AUDIO>); the prefix, the audio positions and <empty> (<EMPTY_AUDIO>) carry no loss.

and tokens due in the same chunk keep their end-time order. Training imposes no limit on the tokens per chunk. On LibriSpeech and KsponSpeech, targets average 0.22 and 0.27 text tokens per chunk, and 75–80% of the chunks emit only <next>; Equation (5) therefore weights <next> by 0.3 (English) or 0.15 (Korean) and every other label, including the planned tags, by 1. The output layer is computed only at labelled positions, which bounds memory on long sequences. Decoding is greedy with a KV cache, with one forward pass per chunk and one per emitted token; it forces <next> after 8 tokens in a chunk or $6K + 64$ tokens in a stream, runs flush rounds until one is empty (at most 8), and blocks the chat-format, audio, transcript-marker, <empty>, speaker and delay tokens. Translation segments will need a higher per-chunk cap.[Draft: planned] On one H200 GPU, decoding with the 0.6B thinker needs 100–150 ms per 80 ms chunk at the 99th percentile, so real time is not yet shown; the 1.7B thinker has not been timed (Table 15). In earlier backbones with the 0.6B thinker, $\delta = 2/3/4$ gave a median emission delay $(k + 1) \cdot 80 \text{ ms} - t$ of about 200/280/360 ms, which excludes the 80 ms encoder look-ahead that Section 5.3 adds, and at most 0.5% of English emissions (2–5% of Korean ones) came 80 ms or more before their target chunk; the evaluation of the 1.7B pilot recorded no emission delays. Figure 11 shows the rule on a 2.6 s utterance: its six words end at 807, 1127, 1447, 1607, 1687 and 1927 ms and are emitted after chunks 12, 16, 20, 22, 23 and 26, each 233 ms after its end; chunks 0–11 and 27–31 emit only <next>, and the flush round is empty.

Training. Stage 0 trains only the adapter, with the thinker frozen, in final mode (the native Qwen3-ASR format), because a randomly initialised adapter otherwise causes very large gradients in the first steps. Stage 1 starts from Stage 0 and the original Qwen3-ASR thinker and trains all parts for one epoch, with $\delta \in \{2, 3, 4, 6, 8\}$ and final mode for each utterance with probability 0.3 (Table 17). Its data are about 3.8k h drawn from a quality-controlled pool of 19 English and Korean sources, plus about 4.3k h of ASR corpora: 806 h of English from LibriSpeech (Panayotov et al., 2015), VoxPopuli (Wang et al., 2021a), Granary-YODAS (Koluguri et al., 2025) and Switchboard (Godfrey et al., 1992), and about 3.5k h of Korean from KsponSpeech (Bang et al., 2020), NIKL conversational speech and AI Hub conversational and broadcast speech; there is no German. We augment with SpecAugment (Park et al., 2019) (2 frequency masks of up to 27 bins and 2 time masks of up to 3% of the utterance), speed perturbation (0.9, 1.0, 1.1), noise (probability 0.4, SNR 5–30 dB), music (SNR 10–30 dB) and reverberation (probability 0.2). The noise bank holds 8.9 h of noise (MUSAN, Snyder et al., 2015; DEMAND, Thiemann et al., 2013; ETSI), 42.6 h of music (MUSAN) and 6,325 simulated and real room impulse responses;

Table 17. Training stages, data and hyperparameters. Stages 0 and 1 exist, and the one-epoch Stage-1 pilot is the backbone of Table 18; Stages 2 and 3 are planned (M1–M3 as in Section 5.4; the decoupled twin of Section 6.5 also starts from M1), and their open settings are blank. Hours are approximate; data sources and augmentation are detailed in Appendix F. lr: peak learning rate after a linear warm-up.

Stage	Trainable	Data	Targets	Schedule	Hardware	Status
0: adapter warm-up	adapter (6.3M); thinker frozen	English and Korean ASR	final mode only (Qwen3-ASR format)	2,000 steps, warm-up 100; 16,384-token batches	1 node	built
1: streaming and final-mode ASR	encoder, adapter, thinker (2,336M)	≈ 3.8 k h curated English and Korean speech + ≈ 4.3 k h English and Korean ASR corpora (806 h English); no German; augmented	lexical transcripts; $\delta \in \{2, 3, 4, 6, 8\}$; final mode with probability 0.3	1 epoch = 3,980 steps; lr 6×10^{-5} thinker (warm-up 500), 10^{-3} adapter, 10^{-5} encoder	2 nodes $\times$ 8 GPUs	built (pilot)
2 (M1): two-speaker bilingual ASR		+ German ASR, 8 kHz telephone-band augmentation, simulated two-speaker En $\leftrightarrow$ De and Ko $\leftrightarrow$ En mono sessions	speaker-tagged transcripts at delay δ_s ; two-slot speaker activity			planned
3 (M2, M3): translation streams		+ CoVoST 2 and MuST-C (En $\leftrightarrow$ De); silver teacher translations in four directions	translations committed at turn ends (M2) or at MU2 (M3), delay δ_t			planned
3 (twin): separate translation pass		as Stage 3	translations of the segments fixed by M1’s speaker tags, at MU2 with delay δ_t			planned

reverberation is applied before noise, and the direct-path peak of each impulse response is aligned to $t = 0$ so that word times stay valid. Stronger time masking (about 12% of the frames) lowered the top-1 accuracy on text tokens at the first step from 0.985 to 0.84, so we avoid it. One epoch takes 3,980 steps on 2 nodes with 8 GPUs each, at about 2.5 s per step including I/O stalls (0.6–1.0 s of compute); the training loss ended at 0.27 and was still falling. The 0.6B pilot uses the same data, steps and augmentation. After training, we reload each checkpoint and compare it with the model in memory key by key, encoder included; for both pilots no key differs, including all 638 encoder tensors. *[Draft: longer Stage-1 runs exist but stay out of the paper until their numbers are recorded with a per-epoch analysis]*

Planned stages. Stage 2 (M1) starts from the 1.7B Stage-1 pilot and adds German ASR data, 8 kHz telephone-band augmentation, the speaker tags and the activity head, trained on simulated two-speaker En $\leftrightarrow$ De and Ko $\leftrightarrow$ En mono sessions whose gaps and overlaps are drawn more broadly than in the benchmark (cf. Landini et al., 2022). Each speaker’s turns are force-aligned before mixing, so that the words of both speakers can be ordered by end time under overlap (Section 5.2). Before training M1, we fixed three acceptance criteria: single-speaker WER and CER at most 5% (relative) above those of the Stage-1 pilot on VoxPopuli-AA and KsponSpeech, cpWER on synthetic L0 sessions at most 1.2 times the matched single-speaker WER, and a speaker-attribution error of at most 5%. Stage 3 then adds the translation streams of Section 5.3, committed at turn ends (M2) or at MU2 (M3), from human En $\leftrightarrow$ De speech translation data, CoVoST 2 (Wang et al., 2021b) and MuST-C (Di Gangi et al., 2019), and from silver teacher translations in four directions. The same-backbone cascade and the decoupled twin of Section 6.5 also start from M1 and train on the same data (Appendix D). Unlike the off-the-shelf baselines, MODEL and both controls thus train on simulated sessions built like the benchmark’s; the fixed-gap sweep (Figure 9) tests whether gains depend on the benchmark’s gap distribution. Training excludes CST-Bench-TAXI, the dialogue corpora and voices of CST-Bench-Syn, and outputs of EXAONE-4.0 (LG AI Research, 2025), which serves only for evaluation (Appendices B and G). *[Draft: planned; German data still has to be obtained, and if M1 misses the criteria twice, M1 transcripts translated by an LLM become our system]*

Evaluation protocol. Backbone results use a single-turn protocol: each utterance is followed by 1 s of digital silence, the state is reset per utterance, and decoding is greedy in fp32 with TF32 disabled. English is scored by WER and Korean by CER without spaces, after normalising the text with numbers spelled out. VoxPopuli-AA is a cleaned release of the VoxPopuli test set (Wang et al., 2021a) (628 utterances, 1.98 h, 292 sessions, 18,072 reference words); AA-WER

Table 18. Streaming ASR backbone on single-speaker English and Korean speech (not CST): error rates in % at decoder delay $\delta = 4$ and $\delta = 6$ and in final mode, encoder context [56, 0], single-turn protocol. WER on English VoxPopuli-AA, together with AA-WER, our approximation of the scoring rules published with this test set; CER without spaces on KsponSpeech eval_clean and eval_other. *Backbone of MODEL. The two Stage-1 models share data, steps and augmentation (Table 17); the earlier 0.6B models are described in Appendix F. Qwen3-ASR-1.7B is non-causal and transcribes whole utterances; the RNN-T is the released Nemotron 3.5 model (the encoder we start from, with its own causal transducer decoder). Both have one operating point. VoxPopuli-AA is not speaker-disjoint from our training data: 80 of its 207 speakers and 280 of its 292 sessions also occur in training. Paired-bootstrap comparisons are given in the text. $\rightarrow$: no final mode.

Model	VoxPopuli-AA WER			VoxPopuli-AA AA-WER			Kspon clean CER			Kspon other CER		
	$\delta = 4$	$\delta = 6$	final	$\delta = 4$	$\delta = 6$	final	$\delta = 4$	$\delta = 6$	final	$\delta = 4$	$\delta = 6$	final
Stage 1, 1.7B thinker*	7.70	7.18	6.84	5.73	5.20	4.77	9.70	9.40	8.21	9.95	9.63	8.42
Stage 1, 0.6B thinker	7.73	7.18	7.14	5.73	5.09	5.09	10.55	10.24	9.27	10.87	10.47	9.22
Earlier recipe, 0.6B	8.94	8.01	—	7.18	6.16	—	12.32	11.76	—	12.86	12.67	—
Earlier + commit tokens, 0.6B	9.88	9.15	—	8.02	7.26	—	12.26	12.04	—	12.26	12.19	—
Qwen3-ASR-1.7B, offline		5.10			3.22			8.92			8.71	
Nemotron RNN-T, [56, 0]		6.88			4.88			19.46			16.80	

approximates the scoring published with it, part of which is not public: we apply Whisper’s text normalisation (Radford et al., 2023), split digits and weight the mean by utterance duration. KsponSpeech (Bang et al., 2020) eval_clean and eval_other have 3,000 and 2,687 utterances. Paired bootstrap tests use 2,000 resamples, clustered by session on VoxPopuli-AA and by utterance on KsponSpeech; we call a difference significant if its 95% interval excludes zero. Differences and relative changes are computed from unrounded error rates.

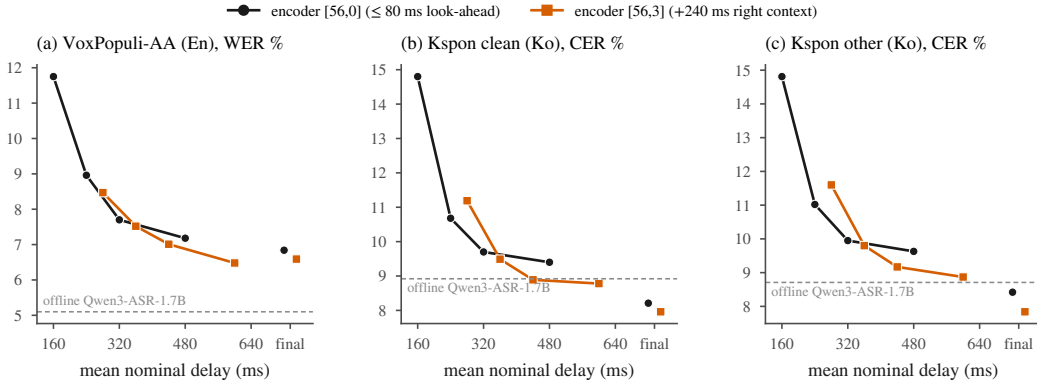


Figure 12. Encoder look-ahead versus decoder delay in the streaming backbone (Stage-1 pilot, 1.7B thinker): error rate against mean nominal delay with encoder context [56, 0] (look-ahead ≤ 80 ms; black circles) and [56, 3] (240 ms more right context; vermilion squares) for $\delta = 2, 3, 4, 6$, and in final mode; dashed: offline Qwen3-ASR-1.7B. (a) VoxPopuli-AA WER (English); (b) KsponSpeech eval_clean and (c) eval_other CER (Korean). The nominal delay is 80δ ms for [56, 0] and $80\delta + 120$ ms for [56, 3]. One checkpoint, trained with sampled encoder contexts, is evaluated at both settings. At matched mean delay, encoder look-ahead and decoder delay are largely interchangeable.

Backbone results. Table 18 compares the two Stage-1 pilots with two earlier 0.6B models and two references. The earlier recipe trained a 0.6B model in several stages on 5.6k h of English and Korean ASR data with $\delta \in \{2, 3, 4, 6\}$ and without final mode; the second earlier model fine-tunes it with sentence-level commit tokens while the encoder is frozen. At $\delta = 4$ and $\delta = 6$, the 1.7B pilot improves on these two models by 10–24% and 19–22% (relative), all significant; only at $\delta = 2$ on KsponSpeech is the commit-token model better, by 0.7–1.0 points. Against the 0.6B pilot, the 1.7B thinker is 6–11% better in every Korean comparison, on par in English streaming, and 4% better in English final mode (6.84 vs. 7.14%; difference -0.31 points, 95% CI $[-0.57, -0.03]$). Against the released Nemotron 3.5 model at [56, 0] (the encoder we start from, with its own causal RNN-T decoder), the English gap closes as δ grows: $+4.87$ points ($+70.8\%$) at $\delta = 2$, $+0.82$ ($+12.0\%$) at $\delta = 4$, $+0.30$ (n.s.) at $\delta = 6$ and -0.04 (n.s.) in final mode. On KsponSpeech, the backbone is 24.0% ($\delta = 2$) to 57.8% (final mode) better on eval_clean and 11.9% to 49.9% better on eval_other. In final mode, it beats offline Qwen3-ASR-1.7B on KsponSpeech eval_clean (8.21 vs. 8.92%; -0.71 points, $[-1.05, -0.37]$), matches it on eval_other

(8.42 vs. 8.71%, n.s.) and trails it in English by 1.73 points (6.84 vs. 5.10%, +33.9%).

Caveats. These are single-speaker results. VoxPopuli-AA is not speaker-disjoint from our training data, because the official VoxPopuli split is not: 80 of its 207 speakers and 280 of its 292 sessions also occur in training, although no utterance or sentence is duplicated. AA-WER only approximates the published scoring. The final-mode gain over offline Qwen3-ASR on KsponSpeech may be an in-domain effect, since the KsponSpeech training set is part of our data. The training data contain no German, and the backbone has not been evaluated on telephone-band CST-Bench-TAXI speech or on two-speaker mixtures, so none of these numbers carries over to CST.

Encoder look-ahead versus decoder delay. Latency can be bought with encoder right context or with decoder delay. As the 1.7B pilot was trained with sampled contexts, we evaluate the same checkpoint at [56, 0] and [56, 3] (Figure 12) and compare the two at their mean nominal delay: 80δ ms for [56, 0] and $80\delta + 120$ ms for [56, 3], whose explicit right context adds 120 ms on average and 240 ms at most. Below, a difference is the error of [56, 3] minus that of [56, 0] in points, given as VoxPopuli-AA WER / KsponSpeech eval_clean CER / eval_other CER, with significance from the paired bootstrap. Between 240 and 320 ms the evidence is mixed: [56, 3] at $\delta = 2$ (280 ms) is better in English than [56, 0] at $\delta = 3$ (240 ms) but worse in Korean ($-0.49 / +0.51 / +0.58$, all significant), and worse everywhere than [56, 0] at $\delta = 4$ (320 ms; $+0.77 / +1.49 / +1.65$, all significant). Between 320 and 480 ms the two are interchangeable: [56, 3] at $\delta = 3$ (360 ms) differs from [56, 0] at $\delta = 4$ (320 ms) by $-0.18 / -0.21 / -0.14$ and from [56, 0] at $\delta = 6$ (480 ms) by $+0.34 / +0.09 / +0.17$, none significant. Right context helps from 440 ms on ([56, 3] at $\delta = 4$ vs. [56, 0] at $\delta = 6$: -0.17 (n.s.) / $-0.51 / -0.46$; at equal $\delta = 6$, i.e. 600 vs. 480 ms: $-0.70 / -0.62 / -0.76$, significant) and in final mode ($-0.25 / -0.25 / -0.58$, all significant), and matching the maximum rather than the mean delay would shrink these gains further. We therefore keep the causal [56, 0] and let δ set the latency, which MODEL conditions separately for each stream.

G. Translation-Unit Pilot

This appendix reports a text-level pilot that motivates the translation commit units of MODEL (Section 5.3). Its conditions differ from those of the benchmark: it simulates simultaneous translation over fixed transcripts with forced-aligned word end times, without recognition in the loop, and the Korean transcripts are themselves ASR output, kept when two recognisers agree; it uses single-speaker Korean and English utterances from monolingual ASR corpora (150 per direction for quality, 300 per language for alignment statistics), not dialogues and not En $\leftrightarrow$ De; Qwen3.8-27B is the incremental translator; the references are offline translations by EXAONE-4.0-32B (LG AI Research, 2025) (silver references); and latency is computation-unaware LAAL on the word times. The pilot compares commit units under these conditions only and does not predict results for speech input, for trained models or on CST-Bench-TAXI.

Commit policies. As in Section 5.3, let $u_{1:I}$ be the transcript of an utterance, e_i the end time of word u_i , and $v = v_{1:|v|}$ the full offline translation by a teacher LLM. A policy commits translation text after it has read b source words, and committed text is never revised, unlike in re-translation (Arivazhagan et al., 2020). We compare four policies. (i) *Utterance end* commits once, after u_I , which amounts to offline translation of the utterance. (ii) *SEM-END* commits at sentence-level semantic-completion boundaries: the positions at which an existing LLM-based labeller, which sees only the source transcript, judges the transcript so far semantically complete, i.e., later words will not change its meaning (in practice mostly the sentence-final punctuation of the transcript); the utterance end is added. These boundaries do not depend on the target language. (iii) *Alignment-only meaning units (MU)*. Contextual alignment (Labiausse et al., 2025) assigns each target word v_j to the source position $\alpha_j = \arg \max_{1 \leq i \leq I} [\phi_j(i) - \phi_j(i-1)]$, where $\phi_j(i)$ is the summed log-probability of the tokens of v_j when v is force-decoded from $u_{1:i}$; ties go to the earliest i . A position i is a *valid cut* if the target words aligned at or before i form a prefix $v_{1:\kappa}$, so that, relative to the teacher’s translation, the output up to that point need not change later. An *MU boundary* is a valid cut that covers more target words than the previous valid cut; I is always a valid cut and an MU boundary. This approximates, with alignment instead of generation, the translation-prefix consistency that defines meaning units (“meaningful units” in Zhang et al., 2020): minimal source segments whose translation is a prefix of the full translation. The policy commits at every MU boundary. (iv) *Verified meaning units (MU2)* commit at a subset of the valid cuts. A candidate is a valid cut $b < I$ that covers more target words than the last commit b' and lies at least a minimum unit after it: 2 Korean eojeol (space-delimited words) or 3 English words, and $e_b - e_{b'} \geq 1.0$ s. At a candidate, the translator receives the source heard so far and the committed translation in its prompt, rather than as a forced prefix, and writes only the new text. A language guard regenerates once and then rejects output in the wrong script: Han or Kana characters, Hangul in English output, or less than 30% Hangul in Korean output. A generation check, in the spirit of Zhang

et al. (2020), accepts the new text only if its sentence-level chrF (Popović, 2015) against the newly covered span $v_{\kappa'+1:\kappa}$ of the teacher’s translation is at least $\theta = 50$, where κ' is the coverage at the last commit. The rest of the utterance is translated at its end.

Setup. We sample 300 utterances per source language with word end times from monolingual Korean and English ASR corpora, plus a second, independent sample of the same size to check sample dependence; the source text is the fixed transcript, and the SEM.END positions are the labeller’s output for it. The labeller has LLM judges rate candidate positions, proposed by an LLM or by rules (e.g., at sentence-final punctuation), and keeps a candidate if its ratings pass thresholds set per candidate type so that precision against LLM-annotated reference labels is at least 0.90 on a development split. Alignments come from Qwen3.8-27B on both samples and from EXAONE-4.0-32B on the first. The quality runs use 150 utterances per direction from the first sample (mean length 11.09 s for Korean and 15.05 s for English sources), with Qwen3.8-27B as translator and as the MU2 teacher. The references are EXAONE-4.0-32B offline translations of the full transcripts, from another model family than the translator; no human references exist for these utterances. Teachers and reference translator receive “*Translate the following $\langle src \rangle$ speech transcript into natural $\langle tgt \rangle$. Output only the $\langle tgt \rangle$ translation, nothing else.*”, followed by the transcript. [Draft: name the source corpora and the sampling arguments, and state whether the English transcripts are human transcripts or, like the Korean ones, agreement-filtered ASR output]

Block A of Table 19 runs the utterance end, SEM.END and MU2 with one incremental prompt and the language guard; only MU2 adds the minimum unit and the generation check. The prompt reads

*You are a simultaneous interpreter translating $\langle src \rangle$ speech into $\langle tgt \rangle$.
Source heard so far: $\langle source prefix \rangle$
Translation already given (cannot be changed): $\langle committed text, or “(none)” \rangle$
Translate only the part of the source that the translation already given does not cover yet, continuing it naturally in $\langle tgt \rangle$.
[The speaker is still talking: translate only what has been said so far and do not guess the rest.] Output only the new $\langle tgt \rangle$ text, nothing else. If nothing new can be translated yet, output nothing.*

where the bracketed sentence appears only before the end of the utterance, and a regeneration by the guard appends “*Write the answer in $\langle tgt \rangle$ only.*” Block B runs the utterance end, SEM.END and alignment-only MUs with the offline prompt, which before the end of the utterance also says “*The transcript may be incomplete because the speaker is still talking: translate only what has been said so far, do not guess or complete the rest.*”; the committed text is forced as the beginning of the answer, and the continuation is committed. Block B has no language guard, and its latency was not reported. In both blocks, the utterance-end translation may use up to 160 tokens and every other step at most $\min(160, 2\nu + 8)$, where ν is the token length of the utterance-end translation, and an m -gram ($m \leq 3$) repeated three or more times in a row is cut.

We report corpus BLEU (Papineni et al., 2002) and chrF with sacreBLEU (Post, 2018), with the BLEU tokenizer `intl` for Korean and `13a` for English output; chrF is primary for Korean output. An output *collapses* if it is empty, is in the wrong script as defined for the guard, or contains an m -gram ($m \leq 3$) repeated three or more times in a row. Latency ignores LLM computation and ASR delay. Output word j gets the commit time $d_j = e_b$ of the step that wrote it, and per utterance

$$\text{LAAL} = \frac{1}{J} \sum_{j=1}^J \left(d_j - (j-1) \frac{e_I}{|Y|} \right),$$

where $|Y|$ is the output length in words and J the first j with $d_j = e_I$ (or $|Y|$ if there is none). This is the length-adaptive form (Papi et al., 2022) of Average Lagging (Ma et al., 2019), except that $|Y|$ replaces $\max(|Y|, |Y^*|)$ with the reference length $|Y^*|$; the two agree whenever the output is at least as long as the reference. *First commit* is d_1 . Both are averaged over utterances, so the LAAL of the utterance end equals the mean utterance length. [Draft: recompute LAAL with $\max(|Y|, |Y^*|)$ when the raw outputs are re-exported] For the alignment statistics, a source word u_i waits $e_{b(i)} - e_i$ until the first boundary $b(i) \geq i$ of a policy.

Verified meaning units halve the lag. In Block A (Table 19 and Figure 13), MU2 roughly halves LAAL relative to the utterance end, from 11.09 to 5.19 s (Ko→En) and from 15.05 to 7.33 s (En→Ko), stays below SEM.END (7.49 and 8.63 s), and makes its first commit earlier (7.03 against 11.09 s and 8.25 against 15.05 s). chrF falls from 61.3 to 60.2 and from 44.2 to 42.0; BLEU falls more, from 35.2 to 31.1 and from 23.4 to 19.9. Whether this reflects paraphrase or errors is untested, since the references are silver and no human judged the outputs. SEM.END scores close to the utterance end, but its output is identical to the utterance-end output in 96/150 (Ko→En) and 83/150 (En→Ko) utterances.

Table 19. Text-level commit-unit pilot (Appendix G). Blocks A and B: Ko↔En, 150 single-speaker utterances per direction, fixed transcripts (Korean: ASR output kept when two recognisers agree) with forced-aligned word end times, Qwen3.8-27B as translator, silver references (offline EXAONE-4.0-32B translations) and computation-unaware LAAL. The two blocks use different prompts, so rows are compared only within a block. Block C (planned): En↔De CST-Bench-TAXI turns with gold transcripts and forced-aligned word times, scored by `cstbench eval` against the human references; its left group is De→En, its right group En→De, and its LAAL is StreamLAAL on the ideal clock. LAAL and First (commit time of the first output word) in seconds. Coll.: collapsed outputs (empty, wrong script or a repetition loop), out of 150 in Blocks A and B. Candidates accepted: MU2 candidates that pass the language guard and the chrF check. -: not reported; blank: pending.

Block / commit policy	Ko→En (C: De→En)					En→Ko (C: En→De)				
	BLEU	chrF	LAAL	First	Coll.	BLEU	chrF	LAAL	First	Coll.
<i>A: prompt-based commits</i>										
Utterance end	35.2	61.3	11.09	11.09	0	23.4	44.2	15.05	15.05	1
SEM.END	35.3	61.1	7.49	8.23	0	22.4	43.7	8.63	9.74	0
Verified MU (MU2)	31.1	60.2	5.19	7.03	0	19.9	42.0	7.33	8.25	1
candidates accepted	197/368 (54%)					185/873 (21%)				
<i>B: forced-prefix continuation (latency not reported)</i>										
Utterance end	32.9	58.4	–	–	–	21.2	42.3	–	–	–
SEM.END	34.4	59.2	–	–	–	19.2	40.8	–	–	–
Alignment-only MU	28.1	54.2	–	–	5	12.9	33.1	–	–	31
<i>C: En↔De CST-Bench-TAXI turns, human references (planned)</i>										
Turn end										
Verified MU (MU2)										
candidates accepted										

SEM.END is sentence-level. Across the three alignment runs, an utterance contains 1.16–1.44 SEM.END units against 3.09–3.45 MUs for Korean sources, and 1.75–1.88 against 5.52–5.73 for English sources. A SEM.END unit spans 11.83–12.82 ejoel against 5.70–6.73 for an MU in Korean, and 24.68–26.15 against 9.22–9.93 words in English (Figure 14). Of all SEM.END positions, 42.3–85.9% coincide with the utterance end, which is a valid cut by definition; counting these, 79.6–92.2% are valid cuts. Yet SEM.END positions cover only 2.0–8.1% of the MU boundaries inside utterances, and a source word waits 3.75–4.05 s (Korean) and 5.20–5.73 s (English) for the next SEM.END boundary, against 2.59–3.25 s and 3.22–3.38 s for the next MU boundary. Semantic completion of the source therefore marks few of the points at which a translation could already be committed. [Draft: also report the valid-cut share of the SEM.END positions inside utterances]

Unverified units collapse, and acceptance depends on the direction. Block B commits without verification: alignment-only MUs continued from a forced prefix collapse in 5/150 Ko→En and 31/150 En→Ko outputs, through text in another script (including copied source), loops or empty output, and their chrF drops to 54.2 and 33.1, against 58.4 and 42.3 for the utterance end of the same block. MU2 collapses in 0/150 and 1/150 utterances; the single En→Ko case is an utterance whose utterance-end output is empty as well. MU2 differs from Block B’s MUs in its prompt, the guard, the minimum unit and the check at once, so the pilot does not separate their effects. The check accepts 197 of 368 candidates (54%) for Ko→En but only 185 of 873 (21%) for En→Ko; nearly all rejections fail the chrF threshold (171 and 685; the other three En→Ko rejections are one empty and two wrong-language outputs). This is consistent with Korean word order: a Korean translation places the verb last, so the translation of an English prefix rarely matches the span of the full translation aligned to it. One threshold may also be stricter for Korean targets, whose chrF is lower overall (44.2 against 61.3 for utterance-end output in Block A). With so few accepted candidates, En→Ko MU2 is only 1.3 s faster than SEM.END, and its output equals the utterance-end output in 52/150 utterances (32/150 for Ko→En).

Extension to En↔De. Block C of Table 19 repeats the comparison on the main pair with human references: CST-Bench-TAXI turns with gold transcripts and forced-aligned word times, Qwen3.8-27B as alignment teacher and translator, $\theta = 50$, a German minimum unit set on held-out data, and trimming of committed text that is re-emitted at the turn end; `cstbench eval` scores the output against the TAXI references. Its MU2 row is the Gold→LLM (MU2) row of Table 3 (non-causal; Appendix D). TAXI turns are much shorter than the pilot utterances (mean source turn 3.086 s in German and 4.624 s in English; mean pilot utterance 11.09 s in Korean and 15.05 s in English), which bounds what commits inside a turn can gain (Figure 10). [Draft: planned; Block C has not been run]

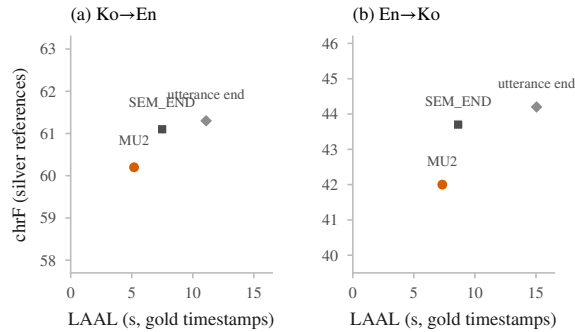


Figure 13. Commit-unit pilot, Block A of Table 19: chrF against computation-unaware LAAL for (a) Ko→En and (b) En→Ko (text level; 150 utterances per direction; fixed transcripts and forced-aligned word times; silver references). Verified meaning units (MU2) roughly halve LAAL relative to committing at the utterance end, at a small chrF cost; SEM_END lies in between on both axes. The LAAL of the utterance end equals the mean utterance length.

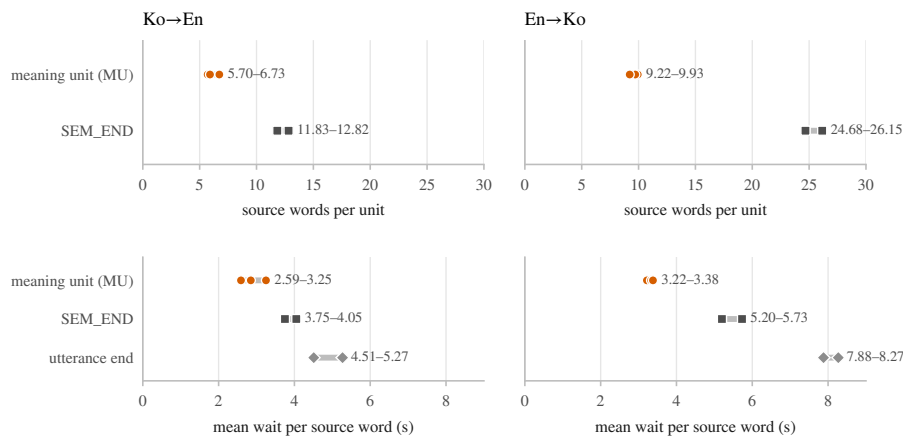


Figure 14. Unit granularity on 300 utterances per source language: source words per unit (Korean: eojeol) and mean wait of a source word until the next commit boundary, for alignment-only meaning units (MU), SEM_END and the utterance end. Dots are three alignment runs (Qwen3.8-27B and EXAONE-4.0-32B on one sample; Qwen3.8-27B on a second sample); bars span the minimum to the maximum. SEM_END and utterance-end values depend on the sample but not on the teacher.

Limitations. The references are silver translations by one LLM, with no human references, COMET or human judgement. The utterances are long, come from single-speaker monolingual corpora and are not dialogues. Transcripts and word times are given rather than recognised (the Korean transcripts being agreement-filtered ASR output), latency ignores computation and ASR delay, and LAAL is computed per utterance rather than as session-level StreamLAAL. There are no confidence intervals: with 150 utterances per direction, small differences, such as between SEM_END and the utterance end, are not interpreted. The translator, the alignment teacher and the target of the generation check are the same model (Qwen3.8-27B), so the check may favour its own phrasing, and neither θ nor the minimum unit was tuned. Finally, MU2 sometimes re-emits committed text at the utterance end; the pilot does not trim it, which may inflate the Ko→En length ratio of MU2 (1.13, against 1.05 for the utterance end).

H. Statistics, Licences and Reproducibility

CST-Bench-TAXI has 86 sessions and about 3.3k reference words per direction, so every result on CST-Bench comes with an uncertainty estimate, and the statistical protocol below is fixed before the test runs. *[Draft: planned: the v0.1 scorer reports no intervals yet; the bootstrap, the Holm correction and the minimum detectable differences are being added; the preliminary runs reported so far predate this protocol and will be re-run under it]*

Session-clustered paired bootstrap. We resample sessions rather than turns, because the turns of one session share speakers, topic and context, and resampling them independently would understate the variance. Each of 2,000 resamples

Table 20. Licences and terms of the corpora, generation models and services behind CST-Bench and MODEL, how we use them, and what we release. Baseline systems are listed in Table 12; scoring, alignment and quality-control models (Appendices B and C) are used unchanged under their own licences and are not redistributed. *Planned: CST-Bench-Syn and the MODEL weights do not exist yet.

Asset	Licence / terms	How we use it	Released by us
BAS TAXI 2.5 (Bavarian Archive for Speech Signals (BAS), 2001; Reichel et al., 2016)	Scientific licence, free of charge (commercial licence available); no redistribution, even in part, without the written permission of the copyright holders	CST-Bench-TAXI, the real test track; evaluation only	Session and turn IDs, configurations, build scripts, checksums, per-turn scores keyed by turn ID; no audio, text or system output
XDailyDialog (Liu et al., 2023)	CC BY-NC-SA 4.0	CST-Bench-Syn dialogues and English–German references; evaluation only	XDailyDialog-based CST-Bench-Syn subset under CC BY-NC-SA 4.0*
BConTrasT (Farajian et al., 2020), built on Taskmaster-1 (Byrne et al., 2019)	CC BY-SA 4.0 (Taskmaster-1: CC BY 4.0)	CST-Bench-Syn dialogues and English–German references; evaluation only	BConTrasT-based CST-Bench-Syn subset under CC BY-SA 4.0*
Nemotron 3.5 ASR Streaming 0.6B (NVIDIA, 2026)	OpenMDW-1.1	Streaming encoder of MODEL, fine-tuned	Encoder within the MODEL weights, under OpenMDW-1.1*
Qwen3-ASR (Shi et al., 2026)	Apache-2.0	Thinker LM of MODEL (1.7B; its audio encoder removed); Qwen3-ASR-1.7B in cascade baselines and for CST-Bench-Syn re-recognition	Thinker within the MODEL weights, under Apache-2.0*
Backbone training corpora (Appendix F): LibriSpeech (Panayotov et al., 2015), VoxPopuli (Wang et al., 2021a), Granary-YODAS, Switchboard, KsponSpeech (Bang et al., 2020), NIKL, AI Hub, a curated pool; MUSAN (Snyder et al., 2015), DEMAND (Thiemann et al., 2013) and ETSI noise	Terms of each corpus (Switchboard: LDC licence)	Training (Stages 0–1) and augmentation of the MODEL backbone; one KsponSpeech transcript is quoted in Figure 11	No data; only the MODEL weights trained on them*
Qwen3.8-27B	Apache-2.0	MT in cascade baselines; alignment, translation and verification teacher of the MU2 commit targets; one of three translators of the CST-Bench-Syn Korean text	Per-turn scores of its runs; none of its translations as references
MADLAD-400-10B-MT (Kudugunta et al., 2023)	Apache-2.0	One of three translators of the CST-Bench-Syn Korean text	Korean text of CST-Bench-Syn*
EXAONE-4.0 (LG AI Research, 2025)	EXAONE AI Model License: non-commercial; outputs may not be used to develop competing models	Internal evaluation only: agreement check of the CST-Bench-Syn Korean text; references and second alignment teacher in the pilot of Appendix G	Nothing; its outputs are neither released nor used for training
Azure neural TTS (Microsoft, 2026)	Service terms of the provider; synthesis on a paid pay-as-you-go resource only, as statements on the free tier conflict	Commercial CST-Bench-Syn edition (main results)	Audio of this edition*
CosyVoice 3 (Du et al., 2025) or Qwen3-TTS (Hu et al., 2026)	Apache-2.0 (weights)	Open CST-Bench-Syn edition on a subset (rank agreement)	Audio of this edition*
cst-bench code and cstbench eval	Apache-2.0	Building CST-Bench-TAXI and CST-Bench-Syn; scoring every system	All code, configurations, checksums and wrappers

draws the sessions of a test set with replacement and keeps every turn of a drawn session. Every metric is recomputed on each resample exactly as on the full test set: BLEU and chrF from the statistics pooled over the resampled turns, COMET over the resampled turns, latency statistics over the resampled non-empty turns, and error rates from pooled counts; per-session scores are never averaged. A 95% interval spans the 2.5th to the 97.5th percentile of the resampled values. A paired comparison scores both systems on the same resamples (Koehn, 2004), and its two-sided p -value is twice the smaller fraction of resampled differences on either side of zero. On CST-Bench-Syn, the resampling unit is the source dialogue, so both speaker–language assignments of a dialogue are drawn together.

Main comparisons and multiple testing. Before the test runs, we fix the main comparisons, the operating point of every system (chosen on the CST-Bench-Syn calibration split; Appendix D) and the decision rules. The main comparisons are oracle against realistic segmentation for each system and direction (RQ1; Section 1), and MODEL against the decoupled twin and against the same-backbone cascade, the two tests of coupling turn-taking and translation (Section 6.5). Holm’s step-down procedure (Holm, 1979) controls the family-wise error rate over them at 0.05. Every other difference is reported with its interval but is not tested.

Rank agreement. Kendall’s τ_b , computed within direction and wrapper strata and averaged (Appendix C), compares system rankings between CST-Bench-TAXI and CST-Bench-Syn, between the two TTS editions, between gold and silver references, across render seeds and between metrics (Table 11). Its 95% interval comes from the same bootstrap: sessions of different test sets are drawn independently, two metrics on one test set share each draw, and τ_b is recomputed on every resample. The decision rule for the agreement between CST-Bench-Syn and CST-Bench-TAXI, a threshold τ^* , is fixed before the test runs, and a low τ_b is reported as a finding.

Minimum detectable difference. For each metric and direction, we compute the smallest paired difference that the test detects at a two-sided level of 0.05 with a power of 0.8, $MDD = (z_{0.975} + z_{0.8}) \hat{\sigma}_\Delta \approx 2.80 \hat{\sigma}_\Delta$, where $\hat{\sigma}_\Delta$ is the session-bootstrap standard error of a paired difference, estimated from the variance of CST-Bench-TAXI segment scores. On CST-Bench-TAXI, the MDDs for En→De and De→En are [TBD] and [TBD] COMET points, [TBD] and [TBD] chrF points, and [TBD] and [TBD] s of median EndOffset. We do not interpret smaller differences and use the same estimate to size CST-Bench-Syn (Appendix B).

Human reference point. Human interpreters are a reference point for our latencies, not a target: their English-to-Korean ear-voice span is about 3 s (Lee, 2002), and we evaluate no human interpretation.

Release. We release one Apache-2.0 package with the CST-Bench-TAXI build scripts, frozen configurations and checksums, the session renderer, `cstbench eval` and the oracle and realistic wrappers of every baseline (Appendix D), together with the per-turn scores of every system keyed by turn ID. TAXI itself is never redistributed. Users obtain the corpus from BAS under its licence (Bavarian Archive for Speech Signals (BAS), 2001; Reichel et al., 2016) and rebuild the track deterministically; Appendix A gives the commands and the verification of timeline digests and audio hashes, and a local rebuild matched 86/86 timeline digests and audio hashes in each L0 configuration. The per-turn scores let anyone who holds the corpus compare new systems with ours without access to our outputs. No TAXI audio, transcript, translation or system output appears in the release or in this paper; apart from one transcript from KsponSpeech (Bang et al., 2020) in Figure 11, all text examples are invented. CST-Bench-Syn is released as two subsets under the licences of their sources, with a data card (Appendix B). The MODEL weights carry both component licences: OpenMDW-1.1 for the encoder and Apache-2.0 for the thinker. [Draft: planned: the realistic wrappers, CST-Bench-Syn and the MODEL weights do not exist yet]

Exact configurations. Every baseline is pinned to a public checkpoint and version (Table 12); the training stages, data and hyperparameters of MODEL are listed in Table 17; render seeds derive from session IDs (Appendix A); and computation-aware latency is measured on one fixed hardware configuration (Table 15). We report sacreBLEU signatures (Post, 2018) and the COMET checkpoint (wmt22-comet-da; Rei et al., 2022a), and we pin every package that changes scores, including the number normaliser. For review, an anonymised copy of the code is available at [TBD]. [Draft: planned: pin the number normaliser and add the anonymised link]

Licences. Table 20 lists the corpora, generation models and services behind CST-Bench and MODEL, their licences or terms, how we use them and what we release. Three constraints shape the release: TAXI may not be redistributed; the two CST-Bench-Syn sources carry incompatible ShareAlike licences, hence two subsets; and the EXAONE AI Model

License restricts EXAONE-4.0 outputs to non-commercial use, so we use them only internally (the agreement check of the CST-Bench-Syn Korean text and the pilot of Appendix G) and neither release them nor train on them. The commercial CST-Bench-Syn edition is synthesised only on a paid Azure resource. No corpus used to train MODEL is redistributed. *[Draft: decision needed: list the licence of each training corpus of Appendix F (the licence of at least one Korean set in the curated pool is unknown) and confirm that these terms allow releasing weights trained on them] [Draft: Table 20: verify every entry, the TAXI copyright holders and the Azure output terms against the current licence texts before submission; decision needed: the licences of the backbone training data (at least one Korean set of the curated pool has an unknown licence), whether they allow releasing the MODEL weights, and whether the KsponSpeech terms allow quoting a transcript]*

I. Extended Related Work

This appendix extends Section 2: it surveys each line of work with its citations, relates `cstbench eval` to existing evaluation tools, and gives the source of every cell of Table 1.

Simultaneous translation policies. A simultaneous system decides after each input step whether to read more input or to write output. The fixed wait- k policy trails the source by k words and comes with Average Lagging (AL; Ma et al., 2019). Learned policies use monotonic attention, from MILk, which also introduced Differentiable Average Lagging (Arivazhagan et al., 2019), to monotonic multihead attention (Ma et al., 2020b); wait- k and monotonic multihead attention were adapted to speech input, together with a computation-aware latency metric (Ma et al., 2020c). Offline models run simultaneously by local agreement, which commits the prefix on which consecutive outputs agree (Liu et al., 2020; Polák et al., 2022; Macháček et al., 2023), or by attention: EDAtt and AlignAtt decide from encoder-decoder attention, or from the alignment derived from it, whether to emit (Papi et al., 2023a;b), and AlignAtt also drives decoder-only LLM translators (Fuxa & Macháček, 2026). Other policies commit at learned meaning units (Zhang et al., 2020) or learn the segmentation of the speech jointly with translation (Zhang & Feng, 2023). Re-translation revises the displayed text instead (Arivazhagan et al., 2020); as CST scores append-only output, such systems take part through a commit rule (Section 6.1).

Streaming systems, LLMs and speech output. StreamAtt extends attention-based policies to unbounded audio by selecting which audio and text history to keep, and introduces StreamLAAL (Papi et al., 2024); a review of 110 SimulST papers finds that most still use human pre-segmented speech (Papi et al., 2025). Speech LLMs translate unbounded speech as a multi-turn dialogue with a managed KV cache (Ouyang et al., 2025b) or with a chunkwise causal encoder (Ouyang et al., 2025a), fold segmentation, policy and translation into one model through speech chain-of-thought (Guo et al., 2025), or act as interpreting agents (Cheng et al., 2024). StreamSpeech, SeamlessStreaming, Hibiki and Hibiki-Zero also emit speech (Zhang et al., 2024; Seamless Communication et al., 2023; Labiausse et al., 2025; 2026): Hibiki processes source and target as synchronous streams, trained with per-word delays from contextual alignment (Labiausse et al., 2025), and Hibiki-Zero dispenses with word-level aligned data by optimising latency with reinforcement learning (Labiausse et al., 2026). All of these systems are built and evaluated for one direction at a time, and none attributes its output to a speaker.

Evaluating simultaneous and streaming translation. SimulEval simulates simultaneous translation of pre-segmented input and reports lagging metrics (Ma et al., 2020a); simulstream evaluates long-form streaming translation, also with re-translation (Gaido et al., 2025). AL underestimates the latency of over-generating systems, which Length-Adaptive Average Lagging (LAAL) corrects (Papi et al., 2022). Stream-level latency resegments the output against the references (Iranzo-Sánchez et al., 2021), as mwerSegmenter does for quality (Matusov et al., 2005), and StreamLAAL applies LAAL to such resegmented output (Papi et al., 2024); a meta-evaluation uncovers a segmentation-related bias of these metrics and introduces YAAL, LongYAAL and the SoftSegmenter resegmenter, implemented in OmniSTEval (Polák et al., 2025). IWSLT moved from sentence-level lagging metrics under a 2-second AL constraint (Ahmad et al., 2024) to unsegmented long-form audio, with StreamLAAL as the only latency metric in 2025 (Abdulmumin et al., 2025) and LongYAAL as the primary one in 2026 (Adelani et al., 2026). All of these tools and metrics score one speaker translated in one direction. `cstbench eval` keeps stream-level resegmentation and StreamLAAL, but resegments each output language onto the turns of its own direction and adds what a listener in a conversation experiences: the wait after each turn (EndOffset, which SimulEval reports per pre-segmented input), the delay at each change of direction (switch latency, SimulEval’s StartOffset taken at such changes) and output in the wrong direction (Section 4 and Appendix C). *[Draft: planned in `cstbench eval`: switch latency and the wrong-direction detector]*

Joint transcription, translation and speaker attribution. Streaming ASR and simultaneous ST can share an encoder, with two synchronized decoders in which intermediate ASR results guide the translation policy (Chen et al., 2021), or one decoder can interleave both, ordered by token-level serialized output training (Papi et al., 2023c); StreamSpeech also emits its streaming transcript (Zhang et al., 2024). Serialized output training (SOT) transcribes overlapping speakers one after another with a single output layer (Kanda et al., 2020b); its token-level form (t-SOT) emits the tokens of overlapping speakers in time order and marks each switch between virtual output channels with a special token (Kanda et al., 2022a), and token-level speaker embeddings attribute these tokens while streaming (Kanda et al., 2022b). End-to-end speaker-attributed ASR also counts and identifies the speakers (Kanda et al., 2020a; 2021). Speaker embeddings let a streaming transducer ST model detect speaker changes, to drive speaker-aware synthesis of the translation (Wang et al., 2025). DiariST builds on t-SOT and token-level speaker embeddings for streaming translation with diarization and scores speaker-attributed BLEU (Yang et al., 2024); STAC-ST emits transcripts, translations and speaker-turn tokens in one serialized output, offline (Zuluaga-Gomez et al., 2023); and JSTAR streams joint ASR and ST with a fast-slow cascaded transducer encoder and multi-objective training (Moritz et al., 2025).

Diarization. End-to-end neural diarization (EEND) maps frames to multi-label speaker activity with a permutation-free objective and thus handles overlap (Fujita et al., 2019a;b); attractors extend it to unknown numbers of speakers (Horiguchi et al., 2020), and a speaker-tracing buffer to streaming input (Xue et al., 2021). Sortformer instead supervises speakers in order of arrival (Park et al., 2025), and Streaming Sortformer keeps the arrival-ordered speaker indices consistent online with a speaker cache (Medennikov et al., 2025). Transducers infer speaker roles (El Shafey et al., 2019) or emit speaker-turn tokens that drive streaming diarization (Xia et al., 2022), while clustering pipelines such as pyannote (Bredin, 2023) and LLM post-correction of speaker labels (Wang et al., 2024) work offline. Two-party benchmarks hold monolingual telephone calls (Godfrey et al., 1992; Cieri et al., 2004; Canavan et al., 1997), meeting benchmarks include single-device recordings (Carletta et al., 2006; Vinnikov et al., 2024), and simulated conversations add controlled overlap (Chen et al., 2020; Cosentino et al., 2020; Landini et al., 2022); none has two speakers who use different languages. Our diarization baselines are pyannote and Streaming Sortformer (Section 6.4), and the activity head of MODEL uses arrival-order slots (Section 5).

Turn-taking and full-duplex models. Conversation analysis describes turns allocated at transition-relevance places with minimal gap and overlap (Sacks et al., 1974), yet measured gaps and overlaps are both frequent (Heldner & Edlund, 2010); computational turn-taking is reviewed by Skantze (2021). Voice Activity Projection predicts the upcoming voice activity of both speakers (Ekstedt & Skantze, 2022) and TurnGPT predicts turn shifts from text (Ekstedt & Skantze, 2020). Monolingual turn-taking models transfer poorly to other languages, and one trained on English, Mandarin and Japanese dialogues matches them (Inoue et al., 2024), but every dialogue it learns from is held in one language. Full-duplex dialogue models represent both sides of a conversation on one timeline, as two audio channels (Nguyen et al., 2023), as parallel audio streams with time-aligned text (Défossez et al., 2024), or by running an LLM synchronously with a real-time clock (Veluri et al., 2024); delayed-stream and step-synchronous decoders emit text on the audio clock (Zeghidour et al., 2025; Seide et al., 2024). MODEL adopts one clock-synchronous token stream for the transcripts and translations of both speakers (Section 5).

Conversational translation systems and data. Machine interpreting of conversations is a long-standing goal (Waibel, 1996; Wahlster, 2000; Lewis, 2015), and JANUS-II already compared push-to-talk with cross-talk recording (Lavie et al., 1996). Appendix A checks public bilingual conversational speech corpora against the requirements of a real CST test set (Table 8): MSLT keeps one side of each bilingual call (Federmann & Lewis, 2016; 2017), DRAL pairs fragments of one-language conversations with re-enactments in the other language (Ward et al., 2023; 2024; Avila & Ward, 2023), InCroMin records meetings mediated by machine translation with one track per participant (Čechovič et al., 2025), the Verbmobil II interpreter volumes need a paid licence (Bavarian Archive for Speech Signals (BAS), 2000), and Fisher-CALLHOME has one source language (Post et al., 2013). Chat translation pairs an English-speaking agent with a German-speaking customer (Farajian et al., 2020), or English with Chinese (Liang et al., 2021), as CST pairs its speakers, but in text. Code-switched speech translation and data (Weller et al., 2022; Alastruey et al., 2023; Hamed et al., 2022; Shankar et al., 2025; Yan et al., 2025; Zhou et al., 2025) alternate languages within a speaker rather than between speakers, which our scope excludes (Section 1); still, one bidirectional end-to-end model translates code-switched speech well even without code-switched training data (Weller et al., 2022), which supports one model for both directions.

Sources of the cells of Table 1. Each cell follows the cited paper; ◦ marks a property that is shown or partly covered but not evaluated, and n.r. one that the paper does not report.

- *SimulST/StreamST*: segmented benchmarks such as MuST-C (Di Gangi et al., 2019) and the long-form IWSLT 2025–2026 test sets (Abdulmumin et al., 2025; Adelani et al., 2026) are public, and some systems release weights.
- *SeamlessStreaming* translates between many languages into text and speech, is evaluated on FLEURS (Conneau et al., 2023) and releases its weights (Seamless Communication et al., 2023).
- *StreamSpeech* translates into English, also emitting its transcript and speech, is evaluated on CVSS (Jia et al., 2022) and releases its weights (Zhang et al., 2024).
- *Hibiki* translates French and *Hibiki-Zero* French, Spanish, Portuguese and German into English, in text and speech; both release weights, and Hibiki-Zero also releases a 45-hour evaluation benchmark (Labiausse et al., 2025; 2026).
- *STAC-ST* is offline; its two-speaker test channels merge the two sides of Spanish Fisher–CALLHOME calls and contain cross-talk (11–12% overlap, marked with cross-talk tokens), and it releases scripts but no models (Zuluaga-Gomez et al., 2023).
- *DiariST* translates multi-speaker Chinese meetings into English with diarization; it is scored on single-channel audio (the first channel of the far-field array, and a mix of the headset signals) with natural overlap, and releases its evaluation set, DiariST-AliMeeting, with offline baselines and the scorer, but not its streaming model (Yang et al., 2024).
- *JSTAR* works on the five-channel microphone array of smart glasses and tells the wearer from the partner by the direction of speech; its single output sequence carries wearer and partner tags, and each speaker’s speech is translated into the other’s language (Spanish–English); it is tested on in-house real conversations and on FLEURS with simulated overlap, and releases neither data nor model (Moritz et al., 2025).
- *Seed LiveInterpret 2.0* depicts a two-speaker Chinese–English conversation (its Fig. 2) and is trained for multi-speaker discrimination, but is evaluated one direction at a time, hence ◦ for speakers × languages and for the choice of direction; its main test set, RealSI, consists of clips cut from public videos, with public annotations and video links (Cheng et al., 2024); the paper does not report its input channels, and the system is closed (Cheng et al., 2025).
- *Ours*: TAXI is free under a scientific licence but may not be redistributed, so we release build scripts and checksums that rebuild CST-Bench-TAXI from its original distribution (Bavarian Archive for Speech Signals (BAS), 2001; Reichel et al., 2016); CST-Bench-Syn is released under CC licences; overlap is rendered (L1, Section 4); and MODEL is released with its weights.

[Draft: planned in our row: the L1 renders, CST-Bench-Syn (incl. Ko↔En), transcript and diarization scoring, and MODEL with its weights]

Use in this paper. Several of these systems serve as baselines (Section 6.1; Appendix D lists checkpoints and wrappers). StreamSpeech and, optionally, Hibiki-Zero, whose weights are public (CC BY-NC-SA 4.0) and which also emits text, run as De→En-only text baselines (Table 13). cstbench eval reuses stream-level resegmentation and StreamLAAL (Papi et al., 2024); MODEL reuses meaning units (Zhang et al., 2020), Hibiki’s contextual alignment (Labiausse et al., 2025), the time-ordered serialization of t-SOT (Kanda et al., 2022a) and arrival-order speakers (Park et al., 2025) (Section 5). As MODEL decides turns, directions and translations in one stream, whether this coupling helps is tested by a decoupled twin and a same-backbone cascade (Section 6.5). [Draft: planned: the baseline runs, Hibiki-Zero among them only as an option, the decoupled twin, and MODEL beyond its streaming ASR backbone]